Supplementary Materials

Full estimates, robustness, earlier analyses, model derivations and exploratory analysis

for **Cheaper Compute, Wider Capability? An Autonomous Synthetic Experiment on Generative AI architecture and Capability Expansion**

Praxis & Youngjin Yoo

y.yoo@lse.ac.uk

The London School of Economics and Political Science

28 October 2026

Repository. https://github.com/yxy23/praxis/blob/main/papers/icis-2026/ICIS-Supplementary-Materials-2026-10-28.pdf

These materials accompany the paper submitted to the ICIS 2026 PDW on Autonomous Scholarship. The paper's workflow appendix (Appendices A to C of the paper) is not repeated here. Sections S-A to S-H and S-K are Appendices A to H and K of the 25 October draft, unchanged in content: section, table and figure labels carry the prefix S, and pointers to the former appendices and to the paper's renumbered displays have been updated. Section S-M holds main-text material moved on 28 October to keep the paper within eight pages; the paper itself now has one table and no figures. Section numbers 6.x–9.x cited inside S-D to S-H refer to the drafts named at the head of each section.

Supplement section	Content	Label in the 25 October draft
S-A	Full estimates, model mapping and multiplicity (Tables S-A1 to S-A5)	Appendix A
S-B	Robustness and missingness (Tables S-B1 to S-B5)	Appendix B
S-C	Routes and retention (Table S-C1)	Appendix C
S-D	The capacity model of 24 October, historical (Tables S-D1 to S-D7)	Appendix D
S-E	The agents' 90-contrast analysis, historical (Tables S-E1 to S-E5)	Appendix E
S-F	The two-factor analysis of exploration purchases, historical (Tables S-F1 to S-F6; Figures S-F1, S-F2)	Appendix F
S-G	Formal model in full (W2.1–W2.11)	Appendix G
S-H	Experimental design and qualification record (W3.1–W3.9)	Appendix H
S-K	Exploratory analysis of exploration over time (Tables S-K1 to S-K9)	Appendix K
S-M.1	Design figure and operationalisation; Figure S-M1, Table S-M1	Section 6.2; Figure 1, Table 1
S-M.2	Estimands and model	Section 6.5
S-M.3	Tables S-M2, S-M4, S-M5, S-M6 and Figures S-M2, S-M3, S-M4	Tables 2, 4, 5, 6; Figures 2, 3, 4
S-M.4	Exploration over time in full; Figure S-M5	Section 8.2; Figure 5

Display in the 25 October draft	In the submission
Table 3	Table 1 of the paper

Display in the 25 October draft	In the submission
Tables 1, 2, 4, 5, 6	Tables S-M1, S-M2, S-M4, S-M5, S-M6
Figures 1 to 5	Figures S-M1 to S-M5
Appendix I; Appendix J	Appendix B; Appendix C of the paper

S-A. Full estimates, model mapping and multiplicity

All values come from the accepted analysis tables of 25 October (Avery, verified run 1; independently reproduced by Scully). B is purchased exploration per round, A is percent of scheduled initially unfamiliar domain-rounds, J_b is percent of worlds with any exploration in rounds 6–30, and a is purchased production assistance per round. Equal world weights and equal branch weights throughout; 106 complete worlds.

S-A.1 Model-to-contrast mapping

The post-period model is $Y_{post} \sim$ Drop × Technology × Propensity with all lower-order terms, propensity as a three-level category and the reference cell integrated, no-shock, low propensity. It is saturated, so its fitted values are the twelve cell means, and each world-level contrast of Section 6.5 is a fixed linear combination of them: the mean of D averages the drop-minus-high differences over technologies; T averages the S − I differences over price branches; X is the S − I difference in drop-minus-high differences; and the joint comparisons contrast these quantities across propensity groups. The before/after model is $Y_{phase} \sim$ Post × Drop × Technology × Propensity, with 24 cells. Because rounds 1–5 are shared, every Drop term without Post is a structural zero, and the substantive shock is the Post × Drop term; the shock-by-technology-by-propensity interaction is the four-way term. Within-world dependence is kept by a pooled unstructured covariance across the four branches (post model) or eight branch-phases (before/after model), estimated from the 106 worlds. The full design matrices, cell means, coefficient covariance matrices and contrast vectors are in `full-model-mapping.json` in the supporting files.

S-A.2 Full model coefficients

Table S-A1. Reference-coded coefficients of the four models (106 complete worlds). Coefficients are effects at the reference categories, not average effects. Structural zeros are fixed at zero by the shared pre-shock history. B and a per round; A in percentage points. Intervals use df = 103.

Model	Term	Estimate	SE	95% CI	Structural zero
B_post	Intercept	+0.0009	0.0005	[−0.0001, 0.0019]	
B_post	Drop	+0.0024	0.0018	[−0.0012, 0.0059]	
B_post	S	0.0000	0.0009	[−0.0017, 0.0017]	
B_post	Medium	−0.0006	0.0007	[−0.0020, 0.0008]	
B_post	High	+0.0002	0.0007	[−0.0012, 0.0017]	
B_post	Drop:S	−0.0003	0.0029	[−0.0060, 0.0054]	
B_post	Drop:Medium	+0.0040	0.0025	[−0.0009, 0.0090]	
B_post	Drop:High	+0.0054	0.0025	[0.0005, 0.0104]	
B_post	S:Medium	+0.0014	0.0012	[−0.0010, 0.0037]	
B_post	S:High	+0.0017	0.0012	[−0.0007, 0.0040]	
B_post	Drop:S:Medium	+0.0017	0.0040	[−0.0062, 0.0096]	
B_post	Drop:S:High	+0.0020	0.0040	[−0.0059, 0.0099]	
B_phase	Intercept	+0.0103	0.0032	[0.0039, 0.0167]	
B_phase	Post	−0.0094	0.0032	[−0.0158, −0.0030]	
B_phase	Drop	0.0000	0.0000	[0.0000, 0.0000]	yes
B_phase	S	−0.0029	0.0051	[−0.0130, 0.0071]	
B_phase	Medium	−0.0047	0.0045	[−0.0137, 0.0042]	
B_phase	High	−0.0006	0.0045	[−0.0095, 0.0084]	
B_phase	Post:Drop	+0.0024	0.0018	[−0.0012, 0.0059]	
B_phase	Post:S	+0.0029	0.0052	[−0.0073, 0.0132]	
B_phase	Post:Medium	+0.0041	0.0045	[−0.0048, 0.0130]	
B_phase	Post:High	+0.0008	0.0045	[−0.0081, 0.0097]	
B_phase	Drop:S	0.0000	0.0000	[0.0000, 0.0000]	yes
B_phase	Drop:Medium	0.0000	0.0000	[0.0000, 0.0000]	yes
B_phase	Drop:High	0.0000	0.0000	[0.0000, 0.0000]	yes
B_phase	S:Medium	+0.0141	0.0071	[0.0000, 0.0281]	
B_phase	S:High	+0.0057	0.0071	[−0.0083, 0.0197]	
B_phase	Post:Drop:S	−0.0003	0.0029	[−0.0060, 0.0054]	
B_phase	Post:Drop:Medium	+0.0040	0.0025	[−0.0009, 0.0090]	

Model	Term	Estimate	SE	95% CI	Structural zero
B_phase	Post:Drop:High	+0.0054	0.0025	[0.0005, 0.0104]	
B_phase	Post:S:Medium	−0.0127	0.0072	[−0.0270, 0.0017]	
B_phase	Post:S:High	−0.0041	0.0072	[−0.0184, 0.0103]	
B_phase	Drop:S:Medium	0.0000	0.0000	[0.0000, 0.0000]	yes
B_phase	Drop:S:High	0.0000	0.0000	[0.0000, 0.0000]	yes
B_phase	Post:Drop:S:Medium	+0.0017	0.0040	[−0.0062, 0.0096]	
B_phase	Post:Drop:S:High	+0.0020	0.0040	[−0.0059, 0.0099]	
A_post	Intercept	+0.57	0.25	[0.08, 1.06]	
A_post	Drop	+0.75	0.37	[0.01, 1.50]	
A_post	S	−0.28	0.31	[−0.91, 0.34]	
A_post	Medium	+0.12	0.34	[−0.56, 0.80]	
A_post	High	−0.35	0.34	[−1.03, 0.33]	
A_post	Drop:S	+0.07	0.64	[−1.21, 1.35]	
A_post	Drop:Medium	−0.13	0.52	[−1.17, 0.90]	
A_post	Drop:High	+0.41	0.52	[−0.62, 1.44]	
A_post	S:Medium	+0.47	0.44	[−0.40, 1.34]	
A_post	S:High	+0.86	0.44	[−0.01, 1.73]	
A_post	Drop:S:Medium	+0.54	0.90	[−1.24, 2.33]	
A_post	Drop:S:High	+0.72	0.90	[−1.06, 2.50]	
A_phase	Intercept	+0.20	0.11	[−0.02, 0.42]	
A_phase	Post	+0.37	0.27	[−0.17, 0.92]	
A_phase	Drop	0.00	0.00	[0.00, 0.00]	yes
A_phase	S	+0.15	0.28	[−0.40, 0.69]	
A_phase	Medium	−0.20	0.15	[−0.50, 0.11]	
A_phase	High	−0.20	0.15	[−0.50, 0.11]	
A_phase	Post:Drop	+0.75	0.37	[0.01, 1.50]	
A_phase	Post:S	−0.43	0.39	[−1.20, 0.34]	
A_phase	Post:Medium	+0.31	0.38	[−0.44, 1.07]	
A_phase	Post:High	−0.15	0.38	[−0.91, 0.61]	
A_phase	Drop:S	0.00	0.00	[0.00, 0.00]	yes
A_phase	Drop:Medium	0.00	0.00	[0.00, 0.00]	yes
A_phase	Drop:High	0.00	0.00	[0.00, 0.00]	yes
A_phase	S:Medium	+0.18	0.38	[−0.58, 0.94]	
A_phase	S:High	+0.32	0.38	[−0.45, 1.08]	
A_phase	Post:Drop:S	+0.07	0.64	[−1.21, 1.35]	
A_phase	Post:Drop:Medium	−0.13	0.52	[−1.17, 0.90]	
A_phase	Post:Drop:High	+0.41	0.52	[−0.62, 1.44]	
A_phase	Post:S:Medium	+0.29	0.54	[−0.78, 1.37]	
A_phase	Post:S:High	+0.54	0.54	[−0.53, 1.62]	
A_phase	Drop:S:Medium	0.00	0.00	[0.00, 0.00]	yes
A_phase	Drop:S:High	0.00	0.00	[0.00, 0.00]	yes
A_phase	Post:Drop:S:Medium	+0.54	0.90	[−1.24, 2.33]	
A_phase	Post:Drop:S:High	+0.72	0.90	[−1.06, 2.50]	

S-A.3 All categorical coefficients

Table S-A2. Medium-minus-low and high-minus-low coefficients of every joint comparison (pooled residual variance, df = 103). The raw p-values are for the single coefficient and belong to no family.

Test	Outcome	Compared quantity	Coefficient	Estimate [95% CI]	p
B03	B	M	medium – low	+0.0025 [0.0007, 0.0044]	.0076
B03	B	M	high – low	+0.0043 [0.0024, 0.0061]	1.2×10^{-5}
B04	B	T	medium – low	+0.0022 [−0.0022, 0.0067]	.321
B04	B	T	high – low	+0.0026 [−0.0018, 0.0071]	.240

Test	Outcome	Compared quantity	Coefficient	Estimate [95% CI]	p
B06	B	D	medium – low	+0.0049 [0.0015, 0.0083]	.0052
B06	B	D	high – low	+0.0064 [0.0030, 0.0098]	2.9×10^{-4}
B07	B	X	medium – low	+0.0017 [−0.0062, 0.0096]	.674
B07	B	X	high – low	+0.0020 [−0.0059, 0.0099]	.624
A03	A	M	medium – low	+0.42 [−0.23, 1.07]	.206
A03	A	M	high – low	+0.47 [−0.19, 1.12]	.159
A04	A	T	medium – low	+0.74 [−0.48, 1.96]	.233
A04	A	T	high – low	+1.22 [−0.01, 2.44]	.051
A06	A	D	medium – low	+0.14 [−0.84, 1.12]	.782
A06	A	D	high – low	+0.77 [−0.21, 1.75]	.122
A07	A	X	medium – low	+0.54 [−1.24, 2.33]	.548
A07	A	X	high – low	+0.72 [−1.06, 2.50]	.426
J02	Jb	M	medium – low	+9.4 [−0.6, 19.5]	.066
J02	Jb	M	high – low	+16.4 [6.3, 26.4]	.0017
J03	Jb	T	medium – low	+15.3 [−4.9, 35.4]	.136
J03	Jb	T	high – low	+12.5 [−7.7, 32.7]	.221

S-A.4 Period means by technology and propensity

Table S-A3. Period means with pointwise 95% world t intervals. Pre-period values are the same history for both price branches. Where every world has the same value, the interval has zero width and no test is defined.

Outcome	Propensity	Technology	Period	Price branch	Mean [95% CI]
B	all	pooled	rounds 1–5	both	0.0104 [0.0076, 0.0132]
B	all	pooled	rounds 6–30	high	0.0013 [0.0007, 0.0018]
B	all	pooled	rounds 6–30	drop	0.0073 [0.0059, 0.0088]
B	all	S	rounds 1–5	both	0.0123 [0.0079, 0.0166]
B	all	S	rounds 6–30	high	0.0018 [0.0010, 0.0026]
B	all	S	rounds 6–30	drop	0.0083 [0.0060, 0.0106]
B	all	I	rounds 1–5	both	0.0085 [0.0049, 0.0121]
B	all	I	rounds 6–30	high	0.0008 [0.0002, 0.0013]
B	all	I	rounds 6–30	drop	0.0063 [0.0042, 0.0084]
B	low	pooled	rounds 1–5	both	0.0088 [0.0041, 0.0136]
B	low	pooled	rounds 6–30	high	0.0009 [0.0001, 0.0017]
B	low	pooled	rounds 6–30	drop	0.0031 [0.0016, 0.0046]
B	low	S	rounds 1–5	both	0.0074 [0.0011, 0.0136]
B	low	S	rounds 6–30	high	0.0009 [−0.0001, 0.0019]

Outcome	Propensity	Technology	Period	Price branch	Mean [95% CI]
B	low	S	rounds 6–30	drop	0.0029 [0.0011, 0.0048]
B	low	I	rounds 1–5	both	0.0103 [0.0031, 0.0175]
B	low	I	rounds 6–30	high	0.0009 [−0.0001, 0.0019]
B	low	I	rounds 6–30	drop	0.0032 [0.0012, 0.0053]
B	medium	pooled	rounds 1–5	both	0.0111 [0.0060, 0.0162]
B	medium	pooled	rounds 6–30	high	0.0010 [0.0003, 0.0017]
B	medium	pooled	rounds 6–30	drop	0.0081 [0.0054, 0.0107]
B	medium	S	rounds 1–5	both	0.0167 [0.0076, 0.0257]
B	medium	S	rounds 6–30	high	0.0017 [0.0004, 0.0029]
B	medium	S	rounds 6–30	drop	0.0094 [0.0057, 0.0132]
B	medium	I	rounds 1–5	both	0.0056 [0.0002, 0.0109]
B	medium	I	rounds 6–30	high	0.0003 [−0.0003, 0.0008]
B	medium	I	rounds 6–30	drop	0.0067 [0.0025, 0.0109]
B	high	pooled	rounds 1–5	both	0.0111 [0.0060, 0.0162]
B	high	pooled	rounds 6–30	high	0.0019 [0.0007, 0.0032]
B	high	pooled	rounds 6–30	drop	0.0106 [0.0079, 0.0132]
B	high	S	rounds 1–5	both	0.0125 [0.0051, 0.0199]
B	high	S	rounds 6–30	high	0.0028 [0.0009, 0.0047]
B	high	S	rounds 6–30	drop	0.0122 [0.0072, 0.0172]
B	high	I	rounds 1–5	both	0.0097 [0.0029, 0.0165]
B	high	I	rounds 6–30	high	0.0011 [−0.0002, 0.0025]
B	high	I	rounds 6–30	drop	0.0089 [0.0047, 0.0131]
A	all	pooled	rounds 1–5	both	0.22 [0.07, 0.37]
A	all	pooled	rounds 6–30	high	0.57 [0.36, 0.79]
A	all	pooled	rounds 6–30	drop	1.67 [1.25, 2.09]
A	all	S	rounds 1–5	both	0.38 [0.10, 0.66]
A	all	S	rounds 6–30	high	0.66 [0.38, 0.94]
A	all	S	rounds 6–30	drop	2.00 [1.36, 2.65]
A	all	I	rounds 1–5	both	0.06 [−0.06, 0.19]
A	all	I	rounds 6–30	high	0.49 [0.21, 0.77]
A	all	I	rounds 6–30	drop	1.34 [0.83, 1.84]
A	low	pooled	rounds 1–5	both	0.27 [−0.04, 0.58]
A	low	pooled	rounds 6–30	high	0.43 [0.11, 0.74]
A	low	pooled	rounds 6–30	drop	1.22 [0.53, 1.91]
A	low	S	rounds 1–5	both	0.34 [−0.15, 0.83]
A	low	S	rounds 6–30	high	0.28 [−0.02, 0.59]
A	low	S	rounds 6–30	drop	1.11 [0.21, 2.00]
A	low	I	rounds 1–5	both	0.20 [−0.20, 0.60]

Outcome	Propensity	Technology	Period	Price branch	Mean [95% CI]
A	low	I	rounds 6–30	high	0.57 [0.06, 1.07]
A	low	I	rounds 6–30	drop	1.32 [0.41, 2.24]
A	medium	pooled	rounds 1–5	both	0.16 [−0.07, 0.39]
A	medium	pooled	rounds 6–30	high	0.78 [0.27, 1.28]
A	medium	pooled	rounds 6–30	drop	1.70 [0.78, 2.63]
A	medium	S	rounds 1–5	both	0.32 [−0.14, 0.79]
A	medium	S	rounds 6–30	high	0.87 [0.25, 1.49]
A	medium	S	rounds 6–30	drop	2.10 [0.76, 3.44]
A	medium	I	rounds 1–5	both	0.00 [0.00, 0.00]
A	medium	I	rounds 6–30	high	0.69 [0.08, 1.29]
A	medium	I	rounds 6–30	drop	1.31 [0.28, 2.33]
A	high	pooled	rounds 1–5	both	0.23 [−0.03, 0.50]
A	high	pooled	rounds 6–30	high	0.51 [0.24, 0.77]
A	high	pooled	rounds 6–30	drop	2.07 [1.52, 2.62]
A	high	S	rounds 1–5	both	0.46 [−0.07, 1.00]
A	high	S	rounds 6–30	high	0.80 [0.32, 1.27]
A	high	S	rounds 6–30	drop	2.75 [1.66, 3.84]
A	high	I	rounds 1–5	both	0.00 [0.00, 0.00]
A	high	I	rounds 6–30	high	0.22 [−0.09, 0.54]
A	high	I	rounds 6–30	drop	1.39 [0.63, 2.14]

S-A.5 All simple effects

Table S-A4. Shock and technology effects within groups. Shock effects are mean difference-in-differences; technology effects are S − I post-period levels averaged over price branches. Paired world t tests; raw p-values belong to no family.

Outcome	Effect	Propensity	Technology	n	Estimate [95% CI]	p
B	technology	low	pooled	34	−0.0001 [−0.0016, 0.0013]	.838
B	shock	low	pooled	34	+0.0022 [0.0005, 0.0039]	.014
B	technology	medium	pooled	36	+0.0021 [−0.0013, 0.0054]	.214
B	shock	medium	pooled	36	+0.0071 [0.0045, 0.0096]	2.1×10^{-6}
B	technology	high	pooled	36	+0.0025 [−0.0016, 0.0066]	.219
B	shock	high	pooled	36	+0.0086 [0.0058, 0.0115]	5.1×10^{-7}
B	shock	all	S	106	+0.0065 [0.0042, 0.0088]	1.2×10^{-7}
B	shock	low	S	34	+0.0021 [0.0000, 0.0041]	.051
B	shock	medium	S	36	+0.0078 [0.0042, 0.0114]	1.1×10^{-4}

Outcome	Effect	Propensity	Technology	n	Estimate [95% CI]	p
B	shock	high	S	36	+0.0094 [0.0042, 0.0147]	8.2×10^{-4}
B	shock	all	I	106	+0.0056 [0.0035, 0.0076]	4.1×10^{-7}
B	shock	low	I	34	+0.0024 [0.0001, 0.0046]	.044
B	shock	medium	I	36	+0.0064 [0.0026, 0.0102]	.0017
B	shock	high	I	36	+0.0078 [0.0036, 0.0120]	6.4×10^{-4}
A	technology	low	pooled	34	−0.25 [−1.04, 0.54]	.526
A	shock	low	pooled	34	+0.79 [0.10, 1.48]	.026
A	technology	medium	pooled	36	+0.49 [−0.44, 1.42]	.291
A	shock	medium	pooled	36	+0.93 [0.08, 1.77]	.033
A	technology	high	pooled	36	+0.97 [0.06, 1.87]	.037
A	shock	high	pooled	36	+1.56 [1.02, 2.10]	1.3×10^{-6}
A	shock	all	S	106	+1.35 [0.71, 1.98]	5.6×10^{-5}
A	shock	low	S	34	+0.82 [−0.11, 1.76]	.082
A	shock	medium	S	36	+1.23 [−0.12, 2.58]	.072
A	shock	high	S	36	+1.95 [0.92, 2.99]	5.0×10^{-4}
A	shock	all	I	106	+0.85 [0.43, 1.27]	1.1×10^{-4}
A	shock	low	I	34	+0.75 [0.06, 1.45]	.035
A	shock	medium	I	36	+0.62 [−0.17, 1.41]	.121
A	shock	high	I	36	+1.17 [0.44, 1.90]	.0026

S-A.6 Paired-world and model-based p-values

Table S-A5. The reproduced paired t test (df = 105) beside the model-based test of the same scalar contrast (df = 103). The marginal contrast variance includes variation between propensity groups; the model uses the pooled within-group residual variance. The difference never changes a conclusion.

Outcome	Contrast	Estimate	Paired t p (df 105)	Model p (df 103)
B	T	+0.0015	.099	.100
B	D	+0.0060	6.4×10^{-13}	5.7×10^{-14}
B	X	+0.0009	.558	.561
A	T	+0.42	.104	.101
A	D	+1.10	3.3×10^{-7}	3.1×10^{-7}
A	X	+0.50	.174	.177

Outcome	Contrast	Estimate	Paired t p (df 105)	Model p (df 103)
Jb	T	+9.4	.025	.024

S-A.7 Multiplicity

The headline family is the fourteen tests of Table 1 of the paper; the supporting family is the seven tests of Table S-M6. Holm adjustment is applied within each family separately. In the headline family, four tests remain below .05 after adjustment: the shock effect on B, the shock effect on A, the propensity level difference in B and the shock-by-propensity interaction in B. The last two overlap (Section 6.5). In the supporting family, two remain below .05: the propensity level difference in participation and the shock effect on production assistance. The historical families of 90, 18 and nine tests are reported with their own adjustments in Sections S-D, S-E and S-F and are not merged with these.

S-B. Robustness and missingness

These displays add no tests and replace none of the results in the main text.

S-B.1 Endowment-standardized estimates

Table S-B1. Primary estimates (equal world weights) beside estimates that give the two endowment records equal weight within each propensity group. The low-propensity group holds 18 record-0 and 16 record-1 worlds; the other groups are balanced, so their values coincide.

Outcome	Quantity	Propensity	Primary	Equal record weights
B	M	low	+0.0020	+0.0020
B	M	medium	+0.0045	+0.0045
B	M	high	+0.0063	+0.0063
B	M	all	+0.0043	+0.0043
B	T	low	−0.0001	−0.0001
B	T	medium	+0.0021	+0.0021
B	T	high	+0.0025	+0.0025
B	T	all	+0.0015	+0.0015
B	D	low	+0.0022	+0.0023
B	D	medium	+0.0071	+0.0071
B	D	high	+0.0086	+0.0086
B	D	all	+0.0060	+0.0061
B	X	low	−0.0003	−0.0004
B	X	medium	+0.0014	+0.0014
B	X	high	+0.0017	+0.0017
B	X	all	+0.0009	+0.0009
A	M	low	+0.82	+0.82
A	M	medium	+1.24	+1.24
A	M	high	+1.29	+1.29
A	M	all	+1.12	+1.12
A	T	low	−0.25	−0.26
A	T	medium	+0.49	+0.49
A	T	high	+0.97	+0.97
A	T	all	+0.42	+0.41
A	D	low	+0.79	+0.80
A	D	medium	+0.93	+0.93
A	D	high	+1.56	+1.56
A	D	all	+1.10	+1.10
A	X	low	+0.07	+0.04
A	X	medium	+0.61	+0.61
A	X	high	+0.79	+0.79
A	X	all	+0.50	+0.49
Jb	M	low	+17.6	+17.9
Jb	M	medium	+27.1	+27.1

Outcome	Quantity	Propensity	Primary	Equal record weights
Jb	M	high	+34.0	+34.0
Jb	M	all	+26.4	+26.5
Jb	T	low	0.0	+0.3
Jb	T	medium	+15.3	+15.3
Jb	T	high	+12.5	+12.5
Jb	T	all	+9.4	+9.5

S-B.2 World-resampled bootstrap intervals

Table S-B2. Percentile intervals from 10,000 whole-world resamples (seed 20261025), stratified by propensity or by the six endowment-by-propensity cells. Each draw keeps all four branches of a world together. Descriptive; not replacement tests.

Strata	Outcome	Quantity	Group	Estimate	95% interval
propensity	B	T	overall	+0.0015	[−0.0002, 0.0033]
propensity	B	T	low	−0.0001	[−0.0016, 0.0012]
propensity	B	T	medium	+0.0021	[−0.0011, 0.0051]
propensity	B	T	medium – low	—	[−0.0013, 0.0056]
propensity	B	T	high	+0.0025	[−0.0014, 0.0064]
propensity	B	T	high – low	—	[−0.0015, 0.0067]
propensity	B	M	overall	+0.0043	[0.0036, 0.0050]
propensity	B	M	low	+0.0020	[0.0012, 0.0028]
propensity	B	M	medium	+0.0045	[0.0031, 0.0060]
propensity	B	M	medium – low	—	[0.0009, 0.0042]
propensity	B	M	high	+0.0063	[0.0049, 0.0076]
propensity	B	M	high – low	—	[0.0026, 0.0059]
propensity	B	D	overall	+0.0060	[0.0048, 0.0074]
propensity	B	D	low	+0.0022	[0.0006, 0.0038]
propensity	B	D	medium	+0.0071	[0.0049, 0.0096]
propensity	B	D	medium – low	—	[0.0020, 0.0078]
propensity	B	D	high	+0.0086	[0.0060, 0.0113]
propensity	B	D	high – low	—	[0.0032, 0.0095]
propensity	B	X	overall	+0.0009	[−0.0023, 0.0041]
propensity	B	X	low	−0.0003	[−0.0029, 0.0021]
propensity	B	X	medium	+0.0014	[−0.0039, 0.0064]
propensity	B	X	medium – low	—	[−0.0043, 0.0073]
propensity	B	X	high	+0.0017	[−0.0056, 0.0092]
propensity	B	X	high – low	—	[−0.0058, 0.0098]
propensity	A	T	overall	+0.42	[−0.08, 0.90]
propensity	A	T	low	−0.25	[−1.05, 0.45]
propensity	A	T	medium	+0.49	[−0.40, 1.37]
propensity	A	T	medium – low	—	[−0.40, 1.90]
propensity	A	T	high	+0.97	[0.13, 1.83]
propensity	A	T	high – low	—	[0.10, 2.40]
propensity	A	M	overall	+1.12	[0.88, 1.39]
propensity	A	M	low	+0.82	[0.46, 1.25]
propensity	A	M	medium	+1.24	[0.73, 1.88]
propensity	A	M	medium – low	—	[−0.26, 1.15]
propensity	A	M	high	+1.29	[0.97, 1.61]
propensity	A	M	high – low	—	[−0.07, 0.95]
propensity	A	D	overall	+1.10	[0.73, 1.50]
propensity	A	D	low	+0.79	[0.20, 1.50]
propensity	A	D	medium	+0.93	[0.24, 1.81]
propensity	A	D	medium – low	—	[−0.89, 1.20]
propensity	A	D	high	+1.56	[1.07, 2.10]
propensity	A	D	high – low	—	[−0.11, 1.56]
propensity	A	X	overall	+0.50	[−0.19, 1.23]
propensity	A	X	low	+0.07	[−0.75, 0.94]

Strata	Outcome	Quantity	Group	Estimate	95% interval
propensity	A	X	medium	+0.61	[−0.64, 2.09]
propensity	A	X	medium – low	—	[−0.98, 2.23]
propensity	A	X	high	+0.79	[−0.57, 2.16]
propensity	A	X	high – low	—	[−0.89, 2.35]
propensity	Jb	T	overall	+9.4	[1.4, 17.5]
propensity	Jb	T	low	0.0	[−11.8, 11.8]
propensity	Jb	T	medium	+15.3	[0.0, 30.6]
propensity	Jb	T	medium – low	—	[−3.3, 34.1]
propensity	Jb	T	high	+12.5	[−1.4, 26.4]
propensity	Jb	T	high – low	—	[−6.0, 31.3]
propensity	Jb	M	overall	+26.4	[22.4, 30.4]
propensity	Jb	M	low	+17.6	[11.0, 25.0]
propensity	Jb	M	medium	+27.1	[20.1, 34.0]
propensity	Jb	M	medium – low	—	[−0.6, 19.4]
propensity	Jb	M	high	+34.0	[27.1, 41.0]
propensity	Jb	M	high – low	—	[6.9, 25.8]
propensity	a	T	overall	+0.0048	[−0.0014, 0.0109]
propensity	a	T	low	+0.0028	[−0.0096, 0.0160]
propensity	a	T	medium	+0.0053	[−0.0038, 0.0144]
propensity	a	T	medium – low	—	[−0.0135, 0.0181]
propensity	a	T	high	+0.0061	[−0.0042, 0.0164]
propensity	a	T	high – low	—	[−0.0131, 0.0197]
propensity	a	M	overall	+0.0304	[0.0273, 0.0335]
propensity	a	M	low	+0.0295	[0.0235, 0.0359]
propensity	a	M	medium	+0.0279	[0.0234, 0.0327]
propensity	a	M	medium – low	—	[−0.0096, 0.0061]
propensity	a	M	high	+0.0336	[0.0284, 0.0390]
propensity	a	M	high – low	—	[−0.0040, 0.0123]
propensity	a	D	overall	+0.0450	[0.0394, 0.0507]
propensity	a	D	low	+0.0443	[0.0338, 0.0557]
propensity	a	D	medium	+0.0378	[0.0296, 0.0464]
propensity	a	D	medium – low	—	[−0.0206, 0.0071]
propensity	a	D	high	+0.0528	[0.0428, 0.0629]
propensity	a	D	high – low	—	[−0.0066, 0.0233]
propensity	a	X	overall	−0.0050	[−0.0178, 0.0082]
propensity	a	X	low	−0.0038	[−0.0300, 0.0250]
propensity	a	X	medium	−0.0078	[−0.0269, 0.0111]
propensity	a	X	medium – low	—	[−0.0383, 0.0287]
propensity	a	X	high	−0.0033	[−0.0242, 0.0181]
propensity	a	X	high – low	—	[−0.0342, 0.0345]
record × propensity	B	T	overall	+0.0015	[−0.0002, 0.0033]
record × propensity	B	T	low	−0.0001	[−0.0016, 0.0012]
record × propensity	B	T	medium	+0.0021	[−0.0012, 0.0050]
record × propensity	B	T	medium – low	—	[−0.0013, 0.0056]
record × propensity	B	T	high	+0.0025	[−0.0013, 0.0063]
record × propensity	B	T	high – low	—	[−0.0014, 0.0067]
record × propensity	B	M	overall	+0.0043	[0.0036, 0.0050]
record × propensity	B	M	low	+0.0020	[0.0013, 0.0027]
record × propensity	B	M	medium	+0.0045	[0.0032, 0.0060]
record × propensity	B	M	medium – low	—	[0.0010, 0.0042]
record × propensity	B	M	high	+0.0063	[0.0049, 0.0076]
record × propensity	B	M	high – low	—	[0.0027, 0.0058]
record × propensity	B	D	overall	+0.0060	[0.0048, 0.0074]
record × propensity	B	D	low	+0.0022	[0.0006, 0.0037]
record × propensity	B	D	medium	+0.0071	[0.0049, 0.0096]
record × propensity	B	D	medium – low	—	[0.0021, 0.0078]
record × propensity	B	D	high	+0.0086	[0.0060, 0.0114]
record × propensity	B	D	high – low	—	[0.0033, 0.0096]
record × propensity	B	X	overall	+0.0009	[−0.0021, 0.0041]
record × propensity	B	X	low	−0.0003	[−0.0029, 0.0021]

Strata	Outcome	Quantity	Group	Estimate	95% interval
record × propensity	B	X	medium	+0.0014	[−0.0039, 0.0064]
record × propensity	B	X	medium – low	—	[−0.0041, 0.0073]
record × propensity	B	X	high	+0.0017	[−0.0056, 0.0089]
record × propensity	B	X	high – low	—	[−0.0057, 0.0094]
record × propensity	A	T	overall	+0.42	[−0.06, 0.88]
record × propensity	A	T	low	−0.25	[−1.05, 0.45]
record × propensity	A	T	medium	+0.49	[−0.39, 1.38]
record × propensity	A	T	medium – low	—	[−0.38, 1.90]
record × propensity	A	T	high	+0.97	[0.14, 1.80]
record × propensity	A	T	high – low	—	[0.12, 2.34]
record × propensity	A	M	overall	+1.12	[0.88, 1.40]
record × propensity	A	M	low	+0.82	[0.47, 1.23]
record × propensity	A	M	medium	+1.24	[0.73, 1.88]
record × propensity	A	M	medium – low	—	[−0.24, 1.14]
record × propensity	A	M	high	+1.29	[0.97, 1.61]
record × propensity	A	M	high – low	—	[−0.05, 0.94]
record × propensity	A	D	overall	+1.10	[0.73, 1.52]
record × propensity	A	D	low	+0.79	[0.20, 1.50]
record × propensity	A	D	medium	+0.93	[0.23, 1.84]
record × propensity	A	D	medium – low	—	[−0.86, 1.21]
record × propensity	A	D	high	+1.56	[1.07, 2.10]
record × propensity	A	D	high – low	—	[−0.09, 1.56]
record × propensity	A	X	overall	+0.50	[−0.16, 1.20]
record × propensity	A	X	low	+0.07	[−0.75, 0.93]
record × propensity	A	X	medium	+0.61	[−0.58, 2.04]
record × propensity	A	X	medium – low	—	[−0.92, 2.15]
record × propensity	A	X	high	+0.79	[−0.50, 2.09]
record × propensity	A	X	high – low	—	[−0.85, 2.28]
record × propensity	Jb	T	overall	+9.4	[1.9, 17.5]
record × propensity	Jb	T	low	0.0	[−11.8, 11.8]
record × propensity	Jb	T	medium	+15.3	[1.4, 29.2]
record × propensity	Jb	T	medium – low	—	[−3.0, 33.4]
record × propensity	Jb	T	high	+12.5	[−1.4, 26.4]
record × propensity	Jb	T	high – low	—	[−6.0, 31.0]
record × propensity	Jb	M	overall	+26.4	[22.6, 30.4]
record × propensity	Jb	M	low	+17.6	[11.0, 25.0]
record × propensity	Jb	M	medium	+27.1	[20.1, 34.0]
record × propensity	Jb	M	medium – low	—	[−0.4, 18.8]
record × propensity	Jb	M	high	+34.0	[27.8, 40.3]
record × propensity	Jb	M	high – low	—	[7.0, 25.5]
record × propensity	a	T	overall	+0.0048	[−0.0014, 0.0107]
record × propensity	a	T	low	+0.0028	[−0.0093, 0.0147]
record × propensity	a	T	medium	+0.0053	[−0.0035, 0.0142]
record × propensity	a	T	medium – low	—	[−0.0128, 0.0175]
record × propensity	a	T	high	+0.0061	[−0.0042, 0.0161]
record × propensity	a	T	high – low	—	[−0.0127, 0.0192]
record × propensity	a	M	overall	+0.0304	[0.0274, 0.0336]
record × propensity	a	M	low	+0.0295	[0.0236, 0.0357]
record × propensity	a	M	medium	+0.0279	[0.0233, 0.0326]
record × propensity	a	M	medium – low	—	[−0.0092, 0.0061]
record × propensity	a	M	high	+0.0336	[0.0285, 0.0389]
record × propensity	a	M	high – low	—	[−0.0040, 0.0119]
record × propensity	a	D	overall	+0.0450	[0.0395, 0.0506]
record × propensity	a	D	low	+0.0443	[0.0341, 0.0549]
record × propensity	a	D	medium	+0.0378	[0.0300, 0.0462]
record × propensity	a	D	medium – low	—	[−0.0197, 0.0067]
record × propensity	a	D	high	+0.0528	[0.0426, 0.0631]
record × propensity	a	D	high – low	—	[−0.0062, 0.0225]
record × propensity	a	X	overall	−0.0050	[−0.0175, 0.0074]
record × propensity	a	X	low	−0.0038	[−0.0285, 0.0209]

Strata	Outcome	Quantity	Group	Estimate	95% interval
record × propensity	a	X	medium	−0.0078	[−0.0267, 0.0108]
record × propensity	a	X	medium – low	—	[−0.0358, 0.0262]
record × propensity	a	X	high	−0.0033	[−0.0239, 0.0178]
record × propensity	a	X	high – low	—	[−0.0323, 0.0325]

S-B.3 Record-adjusted propensity comparisons

Table S-B3. The joint propensity comparisons with the endowment record added as a factor, and the record-by-propensity interaction. Appendix only; unadjusted for multiplicity.

Outcome	Quantity	Propensity F (df)	p	Record × propensity F (df)	p
B	M	11.16 (2,102)	4.1×10^{-5}	0.68 (2,100)	.509
B	T	0.78 (2,102)	.461	0.87 (2,100)	.424
B	D	7.57 (2,102)	8.6×10^{-4}	0.19 (2,100)	.824
B	X	0.14 (2,102)	.866	0.63 (2,100)	.536
A	M	1.19 (2,102)	.307	0.45 (2,100)	.638
A	T	2.01 (2,102)	.139	1.40 (2,100)	.252
A	D	1.39 (2,102)	.254	0.11 (2,100)	.897
A	X	0.36 (2,102)	.702	3.01 (2,100)	.054
Jb	M	5.35 (2,102)	.0062	0.30 (2,100)	.742
Jb	T	1.23 (2,102)	.298	2.56 (2,100)	.082

S-B.4 Scheduled-108 missingness bounds

Table S-B4. Outer bounds over all 108 scheduled worlds, treating every unobserved outcome in the two partial worlds as unknown, never as zero. These are bounds on a finite bank, not confidence intervals. Bounds for B allow any purchase from 0 to 1 in an unobserved round and may be tightened by cash feasibility; bounds for J_b use the common 25-round window. Exact fractions are in the supporting files.

Outcome	Quantity	Propensity	Lower	Upper
A	M	all	+1.10	+1.75
A	M	low	+0.78	+2.72
A	M	medium	+1.24	+1.24
A	M	high	+1.29	+1.29
A	T	all	+0.22	+1.52
A	T	low	−0.79	+3.10
A	T	medium	+0.49	+0.49
A	T	high	+0.97	+0.97
A	D	all	+0.43	+1.73
A	D	low	−1.20	+2.69
A	D	medium	+0.93	+0.93
A	D	high	+1.56	+1.56
A	X	all	−0.81	+1.78
A	X	low	−3.82	+3.95
A	X	medium	+0.61	+0.61
A	X	high	+0.79	+0.79
B	M	all	+0.0042	+0.0107
B	M	low	+0.0019	+0.0214
B	M	medium	+0.0045	+0.0045
B	M	high	+0.0063	+0.0063
B	T	all	−0.0003	+0.0126
B	T	low	−0.0056	+0.0333
B	T	medium	+0.0021	+0.0021
B	T	high	+0.0025	+0.0025

Outcome	Quantity	Propensity	Lower	Upper
B	D	all	−0.0005	+0.0125
B	D	low	−0.0172	+0.0217
B	D	medium	+0.0071	+0.0071
B	D	high	+0.0086	+0.0086
B	X	all	−0.0119	+0.0140
B	X	low	−0.0389	+0.0389
B	X	medium	+0.0014	+0.0014
B	X	high	+0.0017	+0.0017
Jb	T	all	+8.8	+11.1
Jb	T	low	−1.4	+5.6
Jb	T	medium	+15.3	+15.3
Jb	T	high	+12.5	+12.5

S-B.5 Level effects and shock moderation

Because the branches share rounds 1–5, $M = M_{high} + D/2$ and $T = T_{high} + X/2$ hold exactly in every world. Table S-B5 reports the no-shock post-period level and the shared pre-period level, so that the commissioned level effects can be separated from the shock. These comparisons are descriptive decompositions outside both families.

Table S-B5. No-shock and pre-period levels (106 complete worlds).

Outcome	Quantity	Overall mean	p (overall vs 0)	Joint propensity comparison p
B	post-period level, no-shock branch	+0.0013	6.7×10^{-6}	.200
B	S − I, no-shock branch	+0.0010	.034	.329
B	pre-period level	+0.0104	5.1×10^{-11}	.756
A	post-period level, no-shock branch	+0.57	5.3×10^{-7}	.380
A	S − I, no-shock branch	+0.17	.356	.152
A	pre-period level	+0.22	.0046	.845

Because $M = M_{high} + D/2$ holds in every world, the share of a propensity-group difference in the post-period level of exploration that comes from the response to the shock is $(D/2)/M$, taken on the group differences. From the unrounded estimates behind Tables 1 and S-B5 (`categorical-coefficients.csv` and `level-moderation-decomposition.json` in the DiD supporting files of 25 October), the medium-minus-low difference in M is 0.002529 per round, of which $D/2$ is 0.002439, a share of 96.4 percent; the high-minus-low difference is 0.004265, of which $D/2$ is 0.003203, a share of 75.1 percent. The remainders, 0.000090 and 0.001062 per round, are the no-shock level differences whose joint comparison is the p = .200 in the first row of Table S-B5. Section 7.2 reports the shares as about 96 and 75 percent; from the rounded values in Table 1 a reader would obtain 98 and 74 percent.

S-C. Routes and retention

S-C.1 Routes of the shock effect on task expansion

Each successfully served unfamiliar domain-round is assigned to one route in a fixed priority: the worker's own correct answer; otherwise an adopted helper answer in a domain whose card she holds; otherwise an adopted helper answer with no card. The three components sum to A. The priority is an accounting convention, not a causal attribution, and the origin of a held card (purchased analysis, adjacent-input tasks or a helper's method card) is not recorded by it.

Table S-C1. Shock effect on A by route (percentage points; 106 complete worlds; paired t intervals, df = 105).

Route	Shock effect [95% CI]	p
Worker's own correct answer	+0.92 [0.53, 1.31]	8.6×10^{-6}
Adopted helper answer, card held	+0.08 [0.03, 0.13]	.0013
Adopted helper answer, no card (helper only)	+0.10 [0.02, 0.17]	.0089

S-C.2 Retained performance

Retention was measured on separate assessment tasks answered without any live helper or adviser, once with the worker's record and once with the record removed, at rounds 0, 15 and 30. These assessments were analysed in the capacity model of 24 October with equal endowment weights; their tables are preserved as Tables S-D2 and S-D3 and the original family's checkpoint accuracies as Table S-E2. At round 0, record-supported accuracy in initially unfamiliar domains was zero in every cell. At round 30, with her record, the worker answered between 0.52 and 2.11 percent of those assessment tasks correctly; with the record removed, between 0 and 0.07 percent. Across all five domains, record-supported accuracy fell from about 30 percent at round 0 to between 21 and 26 percent at round 30. The empty-record view was certified in advance to be near zero, so its low value is partly a property of the instrument. No assessment withdrew live AI from production on matched tasks. Weak retention therefore neither confirms nor negates the AI-assisted task expansion of the main text, which is a different quantity.

S-D. The capacity model of 24 October (historical)

Preserved from the 24 October draft (SHA256 `0b43d62a…`), with its results unchanged. Appendix, table and figure letters have been renumbered for this paper (old A→D, B→E, C→F, D→G, E→H, F→I); section numbers 6.x–9.x cited inside refer to the 24 October draft (`ICIS Full Draft for Review 2026-10-24.md`).

S-D.1 Specification, inputs and verification

The model, outcome, window, weights, cohorts and nine-test family follow Ada's reviewed specification of 24 October (`capacity-model-realignment/ada-model-spec-reviewed.md`, SHA256 `0bde3e05…`) and its contrast manifest (`ada-outcome-contrast-manifest-reviewed.json`, `85f10e20…`), as reviewed by Esther and Toby and reconciled by Leo. The specification records that it is a dated, author-requested reanalysis after outcome exposure and not a preregistration. Avery implemented it from the six saved input files of the accepted Phase 2 analysis, without new model calls, regrading or repaired source rows, and recorded the specification hash before fitting. The design matrix has exactly the 24 columns endowment × path × [1, T, M, H, T·M, T·H]; a numerical least-squares reconstruction confirms rank 24 and the coefficient-to-contrast mapping. Two final runs gave 28 byte-identical files. Scully independently rederived all 432 world-branch values of A, every cell, contrast, covariance and test, and all 60 figure source rows. One robustness display, the dispatch-time mask, was corrected after that check; the canonical outputs are `capacity-analysis/rob3-correction/corrected-run1` (output manifest `f5f4d92d…`), in which every primary estimate, test and figure source is unchanged. All values below are in percentage points of scheduled unfamiliar domain-rounds unless stated.

Table S-D1. Realized capacity A by endowment (106-world estimates; 108-world bound in parentheses where not point-identified). Record 0 has four unfamiliar domains, record 1 three.

Endowment	Cell	High: estimate (bound)	Drop: estimate (bound)	Pool: estimate (bound)
record 0 (one card)	S, λ = 0	0.39	1.28	0.83
record 0 (one card)	I, λ = 0	0.78	1.17	0.97
record 0 (one card)	S, λ = .5	0.78	1.61	1.19
record 0 (one card)	I, λ = .5	0.33	1.50	0.92
record 0 (one card)	S, λ = .95	1.00	3.50	2.25
record 0 (one card)	I, λ = .95	0.22	0.78	0.50

Endowment	Cell	High: estimate (bound)	Drop: estimate (bound)	Pool: estimate (bound)
record 1 (two cards)	S, λ = 0	0.17 ([0.15, 6.81])	0.92 ([0.81, 7.48])	0.54 ([0.48, 7.15])
record 1 (two cards)	I, λ = 0	0.33 ([0.30, 1.41])	1.50 ([1.33, 2.44])	0.92 ([0.81, 1.93])
record 1 (two cards)	S, λ = .5	0.96	2.59	1.78
record 1 (two cards)	I, λ = .5	1.04	1.11	1.07
record 1 (two cards)	S, λ = .95	0.59	2.00	1.30
record 1 (two cards)	I, λ = .95	0.22	2.00	1.11

S-D.2 Routes

Table S-D2. Route decomposition of A, pooled path (106-world estimates; fixed priority own answer → carded help → helper only). The lower rows give pooled effects by route with 108-world bounds in parentheses; route bounds are joint categorical completions and must not be added at their upper endpoints.

Cell	Own answer	Carded help	Helper only	A
S, λ = 0	0.56	0.06	0.07	0.69
I, λ = 0	0.88	0.01	0.06	0.94
S, λ = .5	1.40	0.04	0.05	1.49
I, λ = .5	0.87	0.03	0.10	1.00
S, λ = .95	1.61	0.08	0.08	1.77
I, λ = .95	0.64	0.05	0.11	0.81
Pooled τ	0.39 ([0.22, 1.51])	0.03 ([−0.16, 1.14])	−0.02 ([−0.21, 1.09])	0.40 ([0.22, 1.52])
Pooled δ(.95)	0.96 (0.96)	0.03 (0.03)	−0.03 (−0.03)	0.97 (0.97)
Pooled γ, .5 − 0	0.42 ([−1.49, 0.45])	0.00 ([−1.95, 0.00])	0.01 ([−1.93, 0.01])	0.42 ([−1.48, 0.47])
Pooled γ, .95 − 0	0.41 ([−1.50, 0.44])	0.03 ([−1.91, 0.03])	0.03 ([−1.91, 0.04])	0.47 ([−1.43, 0.51])
Pooled η, .5	0.84 ([−2.51, 1.38])	−0.03 ([−3.37, 0.52])	−0.06 ([−3.40, 0.49])	0.75 ([−2.61, 1.28])
Pooled η, .95	1.28 ([−2.08, 1.81])	−0.01 ([−3.34, 0.55])	−0.04 ([−3.38, 0.51])	1.22 ([−2.13, 1.76])

S-D.3 Retained performance and earnings

Table S-D3. Assessment without live AI and unfamiliar-domain receipts, pooled path (106-world estimates). L0 columns use only initially unfamiliar domains; "all" columns use all five domains (denominator 40 tasks). R uses the worker's record and no live helper or adviser; U removes the record. Receipts are credits over rounds 6–30, weighted by ρ and the tariff.

Cell	R0, L0	R30, L0 [95%]	U30, L0	R0, all	R30, all	Unfamiliar receipts (credits)
S, λ = 0	0.00	0.52 [0.09, 0.95]	0.00	29.93	23.12	0.38
I, λ = 0	0.00	0.69 [0.15, 1.24]	0.07	29.93	21.02	0.55
S, λ = .5	0.00	1.90 [0.90, 2.89]	0.04	29.86	26.01	1.09
I, λ = .5	0.00	1.22 [−0.07, 2.50]	0.00	29.86	23.47	0.95
S, λ = .95	0.00	2.08 [0.79, 3.38]	0.06	29.93	23.30	1.48
I, λ = .95	0.00	2.11 [0.54, 3.69]	0.06	29.93	24.76	0.59

S-D.4 All effects by path

Table S-D4. Effects on A by path: 106-world estimate, marginal normal 95% interval (unadjusted, uncalibrated) and 108-world bound.

Effect	Path	Estimate	Marginal 95%	108-world bound
τ: S − I, averaged over λ	pool	0.40	[−0.09, 0.89]	[0.22, 1.52]
τ: S − I, averaged over λ	high	0.16	[−0.19, 0.51]	[−0.02, 1.27]
τ: S − I, averaged over λ	drop	0.64	[−0.14, 1.42]	[0.47, 1.76]
δ(0): S − I at λ = 0	pool	−0.26	[−1.04, 0.53]	[−0.79, 3.10]
δ(0): S − I at λ = 0	high	−0.28	[−0.81, 0.26]	[−0.82, 3.06]
δ(0): S − I at λ = 0	drop	−0.24	[−1.40, 0.93]	[−0.76, 3.13]
δ(.5): S − I at λ = .5	pool	0.49	[−0.42, 1.40]	0.49
δ(.5): S − I at λ = .5	high	0.19	[−0.50, 0.87]	0.19
δ(.5): S − I at λ = .5	drop	0.80	[−0.65, 2.24]	0.80
δ(.95): S − I at λ = .95	pool	0.97	[0.12, 1.81]	0.97
δ(.95): S − I at λ = .95	high	0.57	[−0.01, 1.16]	0.57
δ(.95): S − I at λ = .95	drop	1.36	[−0.05, 2.77]	1.36
γ: λ = .5 − λ = 0	pool	0.42	[−0.30, 1.14]	[−1.48, 0.47]
γ: λ = .5 − λ = 0	high	0.36	[−0.21, 0.93]	[−1.57, 0.38]
γ: λ = .5 − λ = 0	drop	0.49	[−0.64, 1.62]	[−1.39, 0.56]
γ: λ = .95 − λ = 0	pool	0.47	[−0.05, 0.99]	[−1.43, 0.51]
γ: λ = .95 − λ = 0	high	0.09	[−0.30, 0.48]	[−1.84, 0.11]
γ: λ = .95 − λ = 0	drop	0.85	[−0.01, 1.72]	[−1.02, 0.92]
Level: λ = .5 − λ = .95	pool	−0.05	[−0.73, 0.63]	−0.05
Level: λ = .5 − λ = .95	high	0.27	[−0.28, 0.82]	0.27
Level: λ = .5 − λ = .95	drop	−0.37	[−1.42, 0.69]	−0.37
η: δ(.5) − δ(0)	pool	0.75	[−0.45, 1.95]	[−2.61, 1.28]
η: δ(.5) − δ(0)	high	0.46	[−0.41, 1.33]	[−2.88, 1.01]
η: δ(.5) − δ(0)	drop	1.03	[−0.82, 2.89]	[−2.33, 1.56]
η: δ(.95) − δ(0)	pool	1.22	[0.07, 2.38]	[−2.13, 1.76]
η: δ(.95) − δ(0)	high	0.85	[0.06, 1.64]	[−2.49, 1.40]
η: δ(.95) − δ(0)	drop	1.60	[−0.23, 3.42]	[−1.77, 2.12]
Middle vs high: δ(.5) − δ(.95)	pool	−0.48	[−1.72, 0.76]	−0.48
Middle vs high: δ(.5) − δ(.95)	high	−0.39	[−1.29, 0.51]	−0.39
Middle vs high: δ(.5) − δ(.95)	drop	−0.56	[−2.58, 1.45]	−0.56

S-D.5 The middle category's package response

Table S-D5. Package response by λ, averaged over designs. The first three rows are differences of the 106-world cell means in Table 1 of the paper; the last two are the specified middle-category readings. Package responses lie outside the two-factor question and compare worlds whose round-5 states already differ by λ.

Reading	Estimate	Marginal 95%	108-world bound
Package response a(drop) − a(high), λ = 0	0.80	—	—
Package response a(drop) − a(high), λ = .5	0.93	—	—
Package response a(drop) − a(high), λ = .95	1.56	—	—
Package response, λ = .5 − λ = 0	0.13	[−0.93, 1.19]	[−1.76, 2.12]
Package response, λ = .5 − λ = .95	−0.63	[−1.62, 0.35]	−0.63

S-D.6 Robustness

Table S-D6. Pooled effects under the fixed robustness displays. The dispatch mask keeps all 108 worlds and treats post-cut observations at positions 99–108 as unknown; its 96-world estimate is descriptive and carries no test.

Pooled effect	Primary estimate	Leave one world out: min to max	Dispatch mask: 108-world bound	Dispatch mask: 96-world estimate
τ: S − I, averaged over λ	0.40	0.31 to 0.49 (106 deletions)	[−9.15, 10.67]	0.32
γ: $\lambda = .5 - \lambda = 0$	0.42	0.21 to 0.54 (106 deletions)	[−9.75, 8.86]	0.55
γ: $\lambda = .95 - \lambda = 0$	0.47	0.41 to 0.59 (106 deletions)	[−9.82, 11.57]	0.50
η: $\delta(.5) - \delta(0)$	0.75	0.47 to 0.93 (106 deletions)	[−19.19, 18.03]	0.94
η: $\delta(.95) - \delta(0)$	1.22	0.95 to 1.39 (106 deletions)	[−21.62, 21.16]	1.35

Table S-D7. Alternative scales and windows, pooled path (106-world estimates).

Cell	A, rounds 6–30	Full window, rounds 1–30	Unfamiliar domains served per round
S, $\lambda = 0$	0.69	0.63	0.025
I, $\lambda = 0$	0.94	0.82	0.033
S, $\lambda = .5$	1.49	1.29	0.051
I, $\lambda = .5$	1.00	0.83	0.034
S, $\lambda = .95$	1.77	1.55	0.064
I, $\lambda = .95$	0.81	0.67	0.027

S-E. The agents' 90-contrast analysis (historical)

Preserved from the 24 October draft (SHA256 `0b43d62a…`), with its results unchanged. Appendix, table and figure letters have been renumbered for this paper (old A→D, B→E, C→F, D→G, E→H, F→I); section numbers 6.x–9.x cited inside refer to the 24 October draft (`ICIS Full Draft for Review 2026-10-24.md`).

This appendix preserves the analysis that the agents' plan of 18 October fixed before launch, with its results as reported on 24 October. It operationalized the author's question through five endpoints, two hypotheses and drop-minus-high contrasts; it is not the capacity model of the main text.

S-E.1 Plan, endpoints and timing

The agents' analysis plan of 18 October defined 90 world-level contrasts: D_S, D_I and Θ for each of the five endpoints in each of the six cases. Each is a case mean over 18 worlds with a standard error of $s/\sqrt{18}$ and 17 degrees of freedom, and the family is adjusted by Holm at .05. The contrasts and their test procedure were fixed and hash-bound before launch, but their inferential status was unresolved, and the reporting plan before launch was descriptive. After the study was complete and the first descriptive analysis had been delivered, the author designated the conditional test procedure active for the 90-member family (23 October). This is a documentary incorporation after collection, not a preregistration. H_1 and H_2 are reported as descriptive companions outside the family, and Θ has no predicted sign.

Table S-E1. The five endpoints of the plan of 18 October

Endpoint	What it counts	Unit
F_{30}	Domains with a correct paid round-30 answer supported by a valid card	domains, 0–5
E_{ext}	Gross receipts over 30 rounds in domains absent at the start, helper credit weighted by ρ	credits
H_{30}	Certification: fresh unaided tasks with the terminal record	accuracy, 0–1
R_{30}	Retained performance: fixed evaluation tasks with the record, no live helper	accuracy, 0–1
U_{30}	The same tasks with an empty record (a diagnostic of text support)	accuracy, 0–1

S-E.2 Accuracy at the checkpoints

Retained performance and certification start near .20 in the one-card endowment and near .40 in the two-card endowment, and they end lower (Table S-E2). They fall from round 0 to round 30 in 44 of 48 measure × case × branch cells. Three more cells, all in record 1, λ = 0, are undetermined by their bounds, and one rises slightly: R in record 1, λ = .5, S drop, from .399 to .408. Accuracy with an empty record (U) is at or near zero everywhere, so the worker shows no hidden-domain competence without its record. Whatever capability the worker has is carried in the text of its record.

Table S-E2. Accuracy with the worker's own record at the checkpoints (all 18 scheduled worlds per case; brackets are missingness bounds)

Case	Branch	R0	R30	H5	H30	U30
reco, λ = 0	S drop	0.199	0.139	0.171	0.139	0.000
reco, λ = 0	S high	0.199	0.124	0.171	0.125	0.000
reco, λ = 0	I drop	0.199	0.149	0.178	0.151	0.000
reco, λ = 0	I high	0.199	0.149	0.178	0.151	0.000
reco, λ = .5	S drop	0.199	0.156	0.172	0.160	0.001
reco, λ = .5	S high	0.199	0.142	0.172	0.146	0.000
reco, λ = .5	I drop	0.199	0.181	0.201	0.183	0.000
reco, λ = .5	I high	0.199	0.144	0.201	0.144	0.000

Case	Branch	R0	R30	H5	H30	U30
rec0, λ = .95	S drop	0.200	0.171	0.192	0.174	0.000
rec0, λ = .95	S high	0.200	0.093	0.192	0.089	0.000
rec0, λ = .95	I drop	0.200	0.154	0.169	0.157	0.000
rec0, λ = .95	I high	0.200	0.111	0.169	0.113	0.000
rec1, λ = 0	S drop	0.397	[0.286, 0.397]	[0.375, 0.431]	[0.283, 0.394]	[0.000, 0.111]
rec1, λ = 0	S high	0.397	[0.303, 0.414]	[0.375, 0.431]	[0.301, 0.412]	[0.000, 0.111]
rec1, λ = 0	I drop	0.397	[0.263, 0.318]	0.343	[0.261, 0.317]	[0.001, 0.057]
rec1, λ = 0	I high	0.397	[0.243, 0.299]	0.343	[0.244, 0.300]	[0.000, 0.056]
rec1, λ = .5	S drop	0.399	0.408	0.388	0.399	0.000
rec1, λ = .5	S high	0.399	0.335	0.388	0.335	0.000
rec1, λ = .5	I drop	0.399	0.281	0.376	0.279	0.000
rec1, λ = .5	I high	0.399	0.333	0.376	0.336	0.000
rec1, λ = .95	S drop	0.399	0.356	0.400	0.350	0.000
rec1, λ = .95	S high	0.399	0.312	0.400	0.314	0.001
rec1, λ = .95	I drop	0.399	0.389	0.399	0.379	0.001
rec1, λ = .95	I high	0.399	0.336	0.399	0.332	0.000

S-E.3 Results

The prespecified family is null. None of the 90 contrasts is rejected by Holm at familywise .05. The smallest raw p is .0018 (F_{30}, record 0, λ = .5, D_I = 4/9 of a domain); its Holm-adjusted p is .164 against a first threshold of .05/90 = .00056. Nine members have raw p < .05 (Table S-E3). Every simultaneous interval includes zero. We do not read these nine as discoveries, and a non-rejection does not establish equality across designs. The complete family is Table S-E5.

Table S-E3. The nine members of the original family with raw p < .05, in Holm order. None is rejected, and every simultaneous interval includes zero. This is not a discovery set.

#	Endpoint	Case	Contrast	Identified	Marginal 95%	Simultaneous	p raw	p Holm
5	F_{30}	rec0, λ = .5	D_I	4/9	[0.190, 0.699]	[−0.066, 0.955]	.002	.164
6	F_{30}	rec0, λ = .5	Θ	−1/2	[−0.852, −0.148]	[−1.206, 0.206]	.008	.717
7	F_{30}	rec0, λ = .95	D_S	1/3	[0.092, 0.575]	[−0.151, 0.818]	.010	.848
34	E_{ext}	rec1, λ = .95	D_S	311/432	[0.192, 1.248]	[−0.340, 1.780]	.010	.907
1	F_{30}	rec0, λ = 0	D_S	−5/18	[−0.507, −0.049]	[−0.738, 0.182]	.020	1
43	H_{30}	rec0, λ = .95	D_S	61/720	[0.014, 0.155]	[−0.057, 0.226]	.021	1
61	R_{30}	rec0, λ = .95	D_S	7/90	[0.007, 0.148]	[−0.063, 0.219]	.032	1
67	R_{30}	rec1, λ = .5	D_S	53/720	[0.006, 0.141]	[−0.062, 0.209]	.034	1
3	F_{30}	rec0, λ = 0	Θ	−2/9	[−0.435, −0.009]	[−0.649, 0.205]	.042	1

Production assistance rose, as H_1 predicts. Across rounds 6–30, 232 of the 300 case-round $D_S(a)$ and $D_I(a)$ cells are positive (108 of 150 for S and 124 of 150 for I), 16 are negative, 27 are zero and 25 straddle zero. The exception is record 1, λ = 0 in the S design, which is above zero in only 4 of 25 rounds. This is directionally consistent with the benchmark and is not a test. Because the high branches are cash-poor and the package includes a rebate, the rise can come from the added liquidity as well as from the lower price.

Receipts rose; service and capability did not grow. The drop branch carries more receipts in initially absent, carded domains (E_{ext}): the 10 point-identified D_S and D_I members are all positive, from 0.17 to 3.16 credits. These amounts are small, and the largest depends heavily on one world. Round-30 paid service is mixed: of 10 point-identified F_{30} members, 6 are positive, 3 negative and 1 zero. For retained performance and certification, 16 of the 20 point-identified $D(R_{30})$ and $D(H_{30})$ members are positive, 2 are zero and 2 negative. Where both branches fall, a positive contrast means that the drop branch eroded less, not that capability grew. Of 192 domain-branch rows with exploration spending, 183 show no measured competence in the explored domain at a later checkpoint (a complete card, $R \geq 6/8$ and $R - U \geq 2/8$), 8 show competence observed after the expenditure and 1 is unknown; those 8 establish only a temporal sequence.

The capability hypothesis: a corrected mapping and a negative companion. H_2 concerns "the one-card, λ = .95 case under the price drop". In the study, one card is record 0. The empirical report of 24 October computed the H_2 companion for record 1, the two-card case, under the same label. Scully verified the endowment coding from the generator, the runtime and all 108 baseline checkpoints, and the companion was recomputed for record 0 without changing anything else. Table S-E4 shows both. The corrected companion covers all 18 one-card, λ = .95 worlds, and all four mean changes are negative. Their medians are zero, and the world signs are mixed. The historical two-card values, also negative, are kept as two-card data and are not H_2.

Table S-E4. The H2 companion: corrected one-card case (record 0, λ = .95, all 18 worlds) beside the historical two-card values

ID	Branch	Change	Mean, one card (corrected)	Worlds + / 0 / −	Historical two-card mean (record 1)
H201	S drop	R30 − R0	−7/240	3 / 11 / 4	−31/720
H202	I drop	R30 − R0	−11/240	1 / 11 / 6	−7/720
H203	S drop	H30 − H5	−13/720	4 / 9 / 5	−1/20
H204	I drop	H30 − H5	−1/80	4 / 10 / 4	−7/360

H_2 was worded "conditional on qualification". No world-level selection was applied, because none was specified. If the phrase refers to the study's pre-launch qualification, which attempt 5 discharged, the companion is H_2 as worded and H_2 is contrary in direction. If it refers to a world-level selection, H_2 as worded was not evaluated. A headroom precondition for the focal case was proposed on 17 October but never adopted or verified; we do not use it to explain the negative result away. The change within a branch also has no counterfactual: time, practice and card turnover are confounded with it.

No ordering of the designs. For Θ, 0 of 30 members are rejected. Raw p < .05 occurs for F_{30} at record 0, λ = 0 (−2/9) and record 0, λ = .5 (−1/2). No ordering of the bundled designs is established.

S-E.4 All 90 members

All 90 members of the prespecified family in family order. "Identified" gives the 18-world case mean, or its bounds where worlds are partial; "outer" marks a conservative outer bound. Marginal 95% and simultaneous intervals are raw and model-conditional ($t_{.975,17} = 2.110$; simultaneous $t = 4.237$). No member is rejected by Holm at .05. Source: `primary-case-bounds90.csv` in the accepted corrected run.

Table S-E5. All 90 members (primary-case-bounds90.csv)

#	End.	Case	Con.	Identified	Bound	Marginal 95%	Simultaneous	p raw	p Holm	Rank
1	F_{30}	rec0, λ = 0	D_S	−5/18	identified	[−0.507, −0.049]	[−0.738, 0.182]	.020	1	5
2	F_{30}	rec0, λ = 0	D_I	−1/18	identified	[−0.173, 0.062]	[−0.291, 0.180]	.331	1	34
3	F_{30}	rec0, λ = 0	Θ	−2/9	identified	[−0.435, −0.009]	[−0.649, 0.205]	.042	1	9
4	F_{30}	rec0, λ = .5	D_S	−1/18	identified	[−0.263, 0.151]	[−0.471, 0.360]	.579	1	52
5	F_{30}	rec0, λ = .5	D_I	4/9	identified	[0.190, 0.699]	[−0.066, 0.955]	.002	.164	1
6	F_{30}	rec0, λ = .5	Θ	−1/2	identified	[−0.852, −0.148]	[−1.206, 0.206]	.008	.717	2
7	F_{30}	rec0, λ = .95	D_S	1/3	identified	[0.092, 0.575]	[−0.151, 0.818]	.010	.848	3
8	F_{30}	rec0, λ = .95	D_I	1/6	identified	[−0.024, 0.357]	[−0.216, 0.550]	.083	1	13
9	F_{30}	rec0, λ = .95	Θ	1/6	identified	[−0.141, 0.474]	[−0.451, 0.784]	.269	1	30
10	F_{30}	rec1, λ = 0	D_S	[−4/9, 2/3]	outer	[−1.284, 1.467]	[−2.130, 2.273]	1	1	63
11	F_{30}	rec1, λ = 0	D_I	[−1/6, 7/18]	outer	[−0.788, 0.983]	[−1.414, 1.582]	1	1	64
12	F_{30}	rec1, λ = 0	Θ	[−5/6, 5/6]	outer	[−2.124, 2.124]	[−3.425, 3.425]	1	1	65
13	F_{30}	rec1, λ = .5	D_S	1/9	identified	[−0.050, 0.272]	[−0.212, 0.434]	.163	1	20
14	F_{30}	rec1, λ = .5	D_I	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	66
15	F_{30}	rec1, λ = .5	Θ	1/9	identified	[−0.050, 0.272]	[−0.212, 0.434]	.163	1	21

#	End.	Case	Con.	Identified	Bound	Marginal 95%	Simultaneous	p raw	p Holm	Rank
16	F_{30}	rec1, λ = .95	D_S	1/6	identified	[−0.024, 0.357]	[−0.216, 0.550]	.083	1	14
17	F_{30}	rec1, λ = .95	D_I	1/9	identified	[−0.123, 0.346]	[−0.360, 0.582]	.331	1	35
18	F_{30}	rec1, λ = .95	Θ	1/18	identified	[−0.213, 0.324]	[−0.483, 0.594]	.668	1	58
19	E_{ext}	rec0, λ = 0	D_S	137/288	identified	[−0.388, 1.339]	[−1.258, 2.209]	.261	1	29
20	E_{ext}	rec0, λ = 0	D_I	13/48	identified	[−0.478, 1.020]	[−1.233, 1.775]	.456	1	46
21	E_{ext}	rec0, λ = 0	Θ	59/288	identified	[−0.368, 0.777]	[−0.945, 1.355]	.461	1	47
22	E_{ext}	rec0, λ = .5	D_S	617/864	identified	[−0.058, 1.486]	[−0.836, 2.265]	.068	1	12
23	E_{ext}	rec0, λ = .5	D_I	1151/576	identified	[−1.614, 5.611]	[−5.256, 9.253]	.259	1	28
24	E_{ext}	rec0, λ = .5	Θ	−2219/1728	identified	[−4.900, 2.332]	[−8.546, 5.978]	.464	1	48
25	E_{ext}	rec0, λ = .95	D_S	5455/1728	identified	[−0.235, 6.549]	[−3.655, 9.969]	.066	1	10
26	E_{ext}	rec0, λ = .95	D_I	1/2	identified	[−0.132, 1.132]	[−0.770, 1.770]	.114	1	16
27	E_{ext}	rec0, λ = .95	Θ	4591/1728	identified	[−0.887, 6.201]	[−4.461, 9.774]	.132	1	18
28	E_{ext}	rec1, λ = 0	D_S	[−767/144, 853/144]	outer	[−15.355, 15.864]	[−25.465, 25.886]	1	1	67
29	E_{ext}	rec1, λ = 0	D_I	[−7/16, 23/16]	outer	[−2.545, 3.425]	[−4.670, 5.430]	1	1	68
30	E_{ext}	rec1, λ = 0	Θ	[−487/72, 229/36]	outer	[−17.194, 16.858]	[−27.710, 27.440]	1	1	69
31	E_{ext}	rec1, λ = .5	D_S	193/144	identified	[−1.324, 4.004]	[−4.010, 6.690]	.303	1	32
32	E_{ext}	rec1, λ = .5	D_I	25/144	identified	[−0.279, 0.626]	[−0.736, 1.083]	.430	1	44
33	E_{ext}	rec1, λ = .5	Θ	7/6	identified	[−1.377, 3.710]	[−3.941, 6.274]	.347	1	40
34	E_{ext}	rec1, λ = .95	D_S	311/432	identified	[0.192, 1.248]	[−0.340, 1.780]	.010	.907	4
35	E_{ext}	rec1, λ = .95	D_I	331/288	identified	[−0.391, 2.689]	[−1.943, 4.242]	.134	1	19
36	E_{ext}	rec1, λ = .95	Θ	−371/864	identified	[−2.175, 1.317]	[−3.936, 3.077]	.611	1	57
37	H_{30}	rec0, λ = 0	D_S	1/72	identified	[−0.040, 0.068]	[−0.094, 0.122]	.593	1	53
38	H_{30}	rec0, λ = 0	D_I	0	identified	[−0.052, 0.052]	[−0.104, 0.104]	1	1	70
39	H_{30}	rec0, λ = 0	Θ	1/72	identified	[−0.058, 0.086]	[−0.131, 0.159]	.690	1	60
40	H_{30}	rec0, λ = .5	D_S	1/72	identified	[−0.042, 0.070]	[−0.098, 0.126]	.607	1	56
41	H_{30}	rec0, λ = .5	D_I	7/180	identified	[−0.020, 0.098]	[−0.079, 0.157]	.181	1	24
42	H_{30}	rec0, λ = .5	Θ	−1/40	identified	[−0.113, 0.063]	[−0.201, 0.151]	.556	1	50
43	H_{30}	rec0, λ = .95	D_S	61/720	identified	[0.014, 0.155]	[−0.057, 0.226]	.021	1	6
44	H_{30}	rec0, λ = .95	D_I	2/45	identified	[−0.024, 0.113]	[−0.094, 0.182]	.190	1	25
45	H_{30}	rec0, λ = .95	Θ	29/720	identified	[−0.066, 0.146]	[−0.173, 0.253]	.434	1	45
46	H_{30}	rec1, λ = 0	D_S	[−31/240, 67/720]	identified	[−0.307, 0.277]	[−0.487, 0.463]	1	1	71
47	H_{30}	rec1, λ = 0	D_I	[−7/180, 13/180]	identified	[−0.207, 0.237]	[−0.376, 0.403]	1	1	72
48	H_{30}	rec1, λ = 0	Θ	[−29/144, 19/144]	identified	[−0.518, 0.420]	[−0.836, 0.710]	1	1	73
49	H_{30}	rec1, λ = .5	D_S	23/360	identified	[−0.005, 0.133]	[−0.074, 0.202]	.067	1	11
50	H_{30}	rec1, λ = .5	D_I	−41/720	identified	[−0.184, 0.070]	[−0.311, 0.197]	.356	1	41
51	H_{30}	rec1, λ = .5	Θ	29/240	identified	[−0.040, 0.282]	[−0.202, 0.444]	.131	1	17
52	H_{30}	rec1, λ = .95	D_S	13/360	identified	[−0.047, 0.120]	[−0.132, 0.204]	.375	1	42
53	H_{30}	rec1, λ = .95	D_I	17/360	identified	[−0.045, 0.139]	[−0.137, 0.231]	.293	1	31
54	H_{30}	rec1, λ = .95	Θ	−1/90	identified	[−0.155, 0.133]	[−0.301, 0.279]	.873	1	61
55	R_{30}	rec0, λ = 0	D_S	11/720	identified	[−0.039, 0.069]	[−0.093, 0.123]	.557	1	51
56	R_{30}	rec0, λ = 0	D_I	0	identified	[−0.052, 0.052]	[−0.103, 0.103]	1	1	74
57	R_{30}	rec0, λ = 0	Θ	11/720	identified	[−0.060, 0.091]	[−0.136, 0.167]	.674	1	59
58	R_{30}	rec0, λ = .5	D_S	1/72	identified	[−0.040, 0.068]	[−0.095, 0.123]	.597	1	54
59	R_{30}	rec0, λ = .5	D_I	13/360	identified	[−0.024, 0.097]	[−0.085, 0.158]	.225	1	26
60	R_{30}	rec0, λ = .5	Θ	−1/45	identified	[−0.110, 0.065]	[−0.198, 0.154]	.599	1	55
61	R_{30}	rec0, λ = .95	D_S	7/90	identified	[0.007, 0.148]	[−0.063, 0.219]	.032	1	7
62	R_{30}	rec0, λ = .95	D_I	31/720	identified	[−0.022, 0.108]	[−0.088, 0.174]	.181	1	23
63	R_{30}	rec0, λ = .95	Θ	5/144	identified	[−0.069, 0.138]	[−0.173, 0.242]	.488	1	49
64	R_{30}	rec1, λ = 0	D_S	[−23/180, 17/180]	identified	[−0.306, 0.278]	[−0.486, 0.464]	1	1	75
65	R_{30}	rec1, λ = 0	D_I	[−13/360, 3/40]	identified	[−0.204, 0.239]	[−0.372, 0.404]	1	1	76
66	R_{30}	rec1, λ = 0	Θ	[−73/360, 47/360]	identified	[−0.517, 0.417]	[−0.834, 0.705]	1	1	77
67	R_{30}	rec1, λ = .5	D_S	53/720	identified	[0.006, 0.141]	[−0.062, 0.209]	.034	1	8
68	R_{30}	rec1, λ = .5	D_I	−19/360	identified	[−0.179, 0.073]	[−0.306, 0.200]	.389	1	43
69	R_{30}	rec1, λ = .5	Θ	91/720	identified	[−0.032, 0.284]	[−0.191, 0.444]	.110	1	15
70	R_{30}	rec1, λ = .95	D_S	31/720	identified	[−0.044, 0.131]	[−0.133, 0.219]	.314	1	33
71	R_{30}	rec1, λ = .95	D_I	19/360	identified	[−0.042, 0.148]	[−0.138, 0.243]	.256	1	27

#	End.	Case	Con.	Identified	Bound	Marginal 95%	Simultaneous	p raw	p Holm	Rank
72	R_{30}	rec1, λ = .95	Θ	−7/720	identified	[−0.162, 0.142]	[−0.315, 0.296]	.894	1	62
73	U_{30}	rec0, λ = 0	D_S	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	78
74	U_{30}	rec0, λ = 0	D_I	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	79
75	U_{30}	rec0, λ = 0	Θ	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	80
76	U_{30}	rec0, λ = .5	D_S	1/720	identified	[−0.002, 0.004]	[−0.004, 0.007]	.331	1	36
77	U_{30}	rec0, λ = .5	D_I	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	81
78	U_{30}	rec0, λ = .5	Θ	1/720	identified	[−0.002, 0.004]	[−0.004, 0.007]	.331	1	37
79	U_{30}	rec0, λ = .95	D_S	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	82
80	U_{30}	rec0, λ = .95	D_I	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	83
81	U_{30}	rec0, λ = .95	Θ	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	84
82	U_{30}	rec1, λ = 0	D_S	[−1/9, 1/9]	identified	[−0.272, 0.272]	[−0.434, 0.434]	1	1	85
83	U_{30}	rec1, λ = 0	D_I	[−13/240, 41/720]	identified	[−0.172, 0.174]	[−0.290, 0.292]	1	1	86
84	U_{30}	rec1, λ = 0	Θ	[−121/720, 119/720]	identified	[−0.424, 0.421]	[−0.681, 0.680]	1	1	87
85	U_{30}	rec1, λ = .5	D_S	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	88
86	U_{30}	rec1, λ = .5	D_I	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	89
87	U_{30}	rec1, λ = .5	Θ	0	identified	[0.000, 0.000]	[0.000, 0.000]	1	1	90
88	U_{30}	rec1, λ = .95	D_S	−1/720	identified	[−0.004, 0.002]	[−0.007, 0.004]	.331	1	38
89	U_{30}	rec1, λ = .95	D_I	1/720	identified	[−0.002, 0.004]	[−0.004, 0.007]	.331	1	39
90	U_{30}	rec1, λ = .95	Θ	−1/360	identified	[−0.007, 0.001]	[−0.011, 0.005]	.163	1	22

S-F. The two-factor analysis of exploration purchases (historical)

Preserved from the 24 October draft (SHA256 `0b43d62a…`), with its results unchanged. Appendix, table and figure letters have been renumbered for this paper (old A→D, B→E, C→F, D→G, E→H, F→I); section numbers 6.x–9.x cited inside refer to the 24 October draft (`ICIS Full Draft for Review 2026-10-24.md`).

This appendix preserves the two-factor analysis that the author requested on 24 October and that the agents specified on exploration purchases after the outcomes. Its outcomes are supporting behavior, not capacity. Its findings, including the corrected one-card mapping and the claims corrections made before this draft, are reported as they were verified.

S-F.1 Specification

The additional two-factor analysis. On 24 October the author asked for "Exactly the two way main effects, interactions for the impact of propensity and technology." This analysis was specified after the outcomes. We label it "additional theory-motivated analyses of previously articulated expectations, not specified in the final SAP." The author chose exploration quantity and any exploration over rounds 6–30 as the outcomes for the central question; extending the window to other outcomes, and the statistical method itself, are agent choices.

For $Y \in \{B, J_b\}$, let $m_{r,\ell,d,p}$ be the mean of Y over the 18 scheduled worlds with record r and weight ℓ, in design d and price path p. The pooled path averages the two paths within each world before aggregation, so it gives equal weight to the no-shock path without rebate and to the price-plus-rebate package; it is not a third experiment. The six cells are $\mu_{d,\ell,p} = (m_{0,\ell,d,p} + m_{1,\ell,d,p})/2$, which adjusts for the endowment by equal weights. With $\delta_{\ell,p} = \mu_{S,\ell,p} - \mu_{I,\ell,p}$ and $a_{\ell,p} = (\mu_{S,\ell,p} + \mu_{I,\ell,p})/2$, the three effects are

$$\tau_p = \tfrac{1}{3}\left(\delta_{0,p} + \delta_{.5,p} + \delta_{.95,p}\right),$$
$$\gamma_p = \left(a_{.5,p} - a_{0,p},\ a_{.95,p} - a_{0,p}\right),$$
$$\eta_p = \left(\delta_{.5,p} - \delta_{0,p},\ \delta_{.95,p} - \delta_{0,p}\right).$$

These are marginal contrasts in the saturated six-cell representation. A main effect averaged over the other factor is not a constant effect at every level, and no interaction is assumed away to estimate the main effects.

Two kinds of evidence answer each question, and we report both every time. The primary evidence is the finite bank: each cell and contrast over all 108 scheduled worlds, with missingness bounds. The conditional evidence is a set of exactly 18 two-sided Wald tests, one for each effect, outcome and price path, computed on the 106 complete-vector worlds with a χ^2 reference (1 degree of freedom for τ, 2 for γ and η) and adjusted by Holm across the 18. They assume that worlds are independent and identically distributed within each case, that the provider is stationary, that the complete-vector selection is ignorable, and that the asymptotic approximation is adequate. Different-λ worlds are never paired. These tests are approximate and potentially liberal with only 16–18 complete worlds per case. The false-positive rate, direction and magnitude of approximation error have not been calibrated for this heterogeneous, potentially dependent bank. Holm adjustment does not repair invalid marginal p-values or post-outcome analytical selection. Marginal normal intervals are unadjusted displays and create no additional rejection or equality claims. A non-rejection does not show that an effect is zero, and a rejection on one path does not show an effect on both. The 90-member and 18-member families are separate; we make no combined confirmatory claim.

Descriptive companions accompany the tests: drop-minus-high responses in assistance and exploration, their case-level values, and capability changes. For each we report the distribution across worlds (positive, zero, negative and sign-ambiguous counts), because a mean can hide opposite signs. Robustness analyses repeat every display on expression-complete cohorts, by endowment, with each world left out in turn, and at the dispatch-time cut. They add no tests.

S-F.2 Technology and propensity in exploration

This section reports the two-factor analysis of exploration purchases as delivered on 24 October. Its results are post-outcome and conditional, and they do not test the propositions of Section S-G (W2.9). Table S-F1 and Figures S-F1 and S-F2 show all six cells on both price paths and in the pooled average. In each cell, the 108-world bounds come first and the 106-world complete-vector estimate second.

Table S-F1. The six cells for exploration quantity B and any exploration J_b. For each path: 108-world bounds (a single value when point-identified), then the 106-world complete-vector estimate. Pool averages, within each world, the no-shock path without rebate and the price-plus-rebate package; it is not a third experiment.

Y	λ	Design	Drop: bounds	Drop: est.	High: bounds	High: est.	Pool: bounds	Pool: est.
B	0	S	[0.076, 0.910]	0.076	[0.021, 0.854]	0.023	[0.049, 0.882]	0.049
B	0	I	[0.076, 0.215]	0.083	[0.021, 0.160]	0.021	[0.049, 0.188]	0.052
B	.5	S	0.236	0.236	0.042	0.042	0.139	0.139
B	.5	I	0.167	0.167	0.007	0.007	0.087	0.087
B	.95	S	0.306	0.306	0.069	0.069	0.188	0.188
B	.95	I	0.222	0.222	0.028	0.028	0.125	0.125
J_b	0	S	[0.278, 0.306]	0.271	[0.083, 0.139]	0.090	[0.181, 0.222]	0.181
J_b	0	I	[0.250, 0.278]	0.271	[0.083, 0.111]	0.083	[0.167, 0.194]	0.177
J_b	.5	S	0.528	0.528	0.167	0.167	0.347	0.347
J_b	.5	I	0.361	0.361	0.028	0.028	0.194	0.194
J_b	.95	S	0.583	0.583	0.222	0.222	0.403	0.403
J_b	.95	I	0.472	0.472	0.083	0.083	0.278	0.278

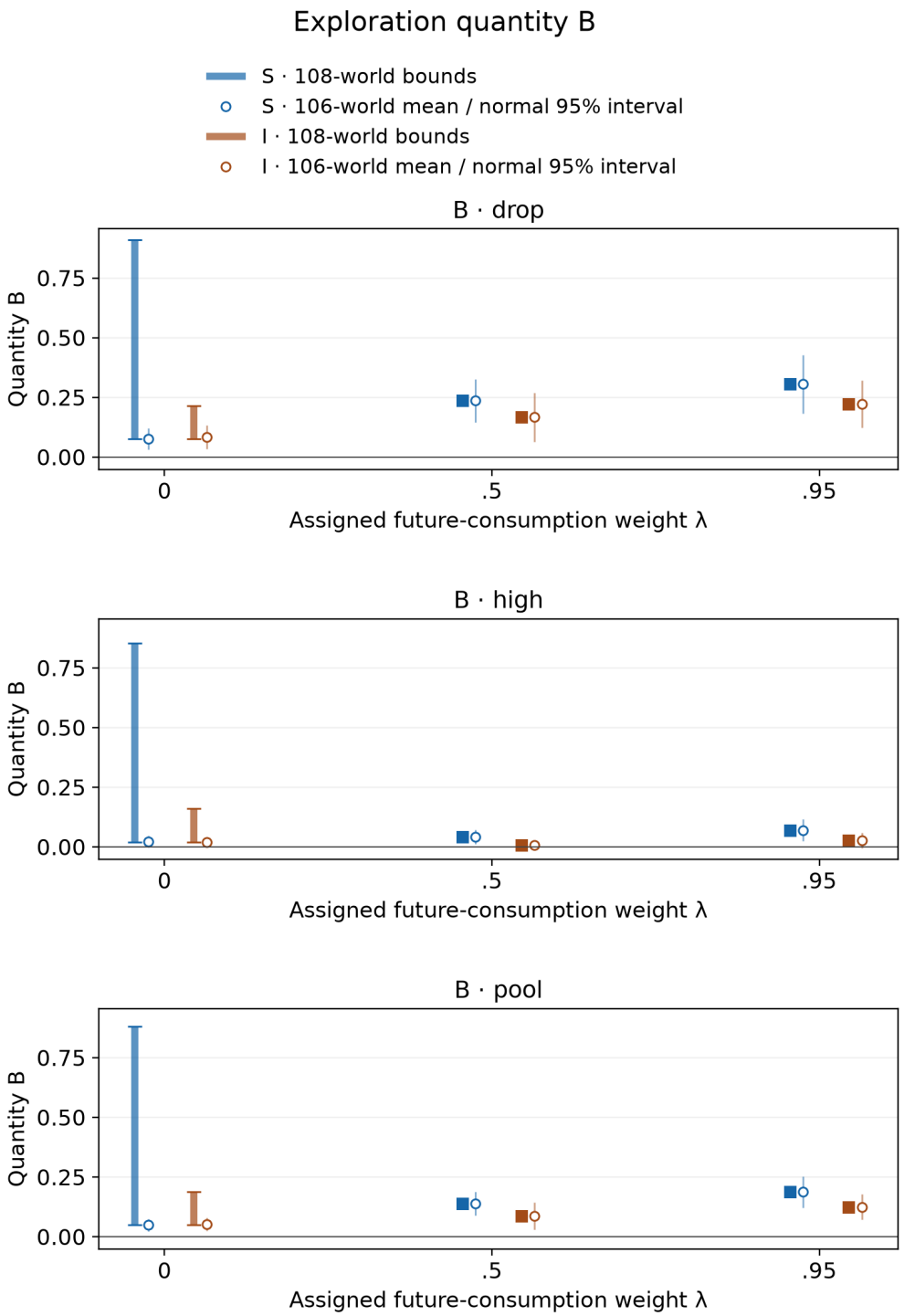


Figure S-F1. Exploration quantity B in the six cells, by price path. Thick bars with caps are 108-world missingness bounds; squares mark point-identified values. Open circles with thin bars are 106-world complete-vector means with unadjusted marginal normal 95% intervals, which create no additional rejection claims. The pool panel averages, within each world, the no-shock path without rebate and the price-plus-rebate package.

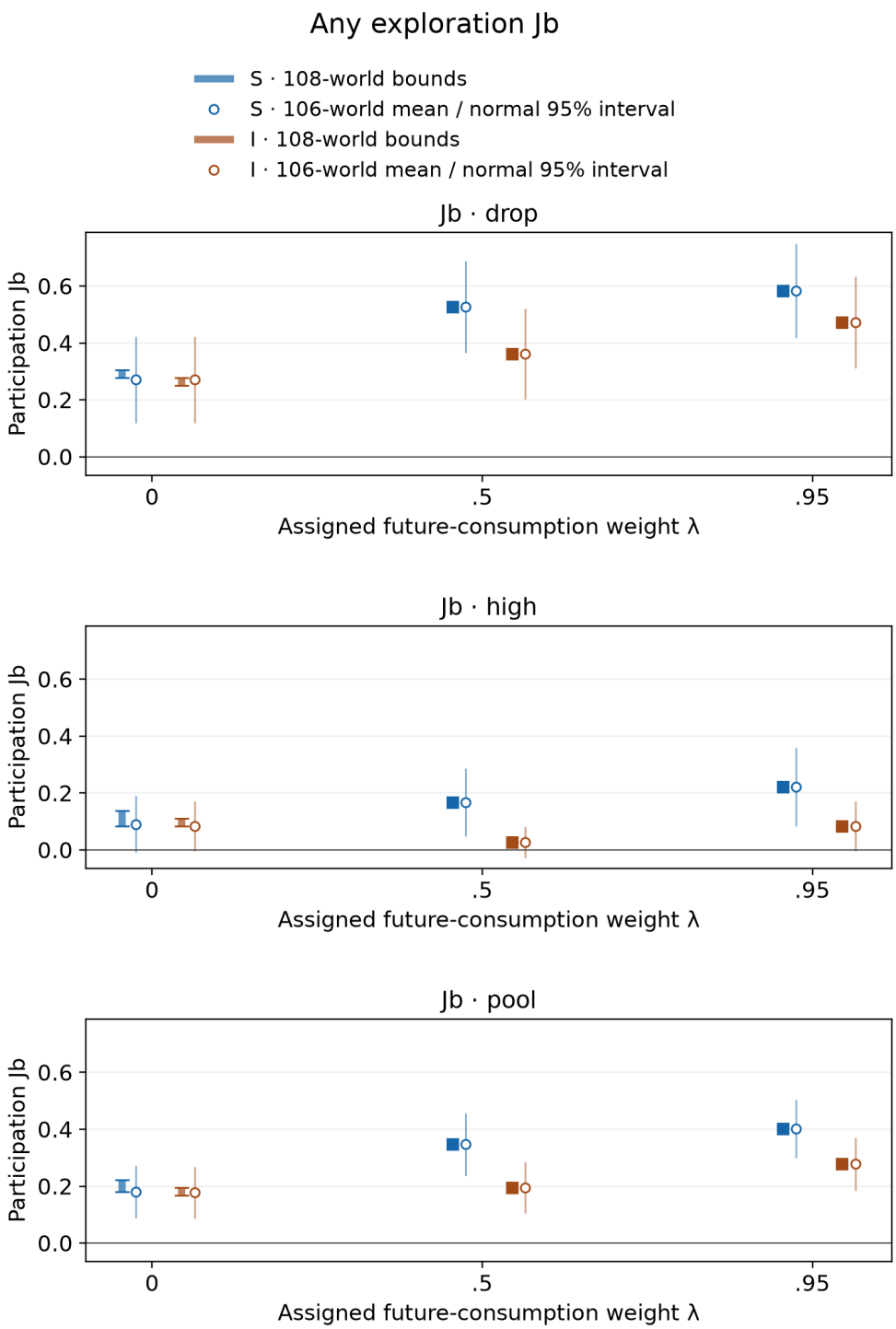


Figure S-F2. Any exploration J_b in the six cells, by price path. Symbols as in Figure S-F1. J_b is the share of world branches with any exploration purchase in rounds 6–30.

The λ = 0 benchmark comes first. At λ = 0, an exact maximiser never explores. The workers did. At least 18 of the 36 λ = 0 worlds explored in rounds 6–30 (7 of 18 in record 0 and 11 of 18 in record 1); 17 did not, and 1 is sign-ambiguous because of unobserved rounds. The 108-world bound on λ = 0 exploration quantity is 7/144 to 77/144, and the pooled λ = 0 cells have lower bounds of 7/144 in both designs for B and of 13/72 (S)

and 1/6 (I) for J_b. This is the first departure from the benchmark, and it shapes how the two-factor contrasts read: the λ = 0 cells are not zero, so the contrasts against λ = 0 are not contrasts against a benchmark of no exploration. It does not show a human trait or irrationality; λ = 0 workers responded to an administered instruction.

The magnitudes are small throughout. A full exploration purchase is $b = 1$ in one round, and B sums purchases over 25 rounds. The largest cell, S at λ = .95 on the drop path, averages 11/36, less than a third of one full purchase per branch. Even there, 7/12 of world branches explored at all.

Table S-F2. The 18 conditional sensitivity tests (106 complete worlds; Holm across 18) beside the 108-world bounds of each tested component. Pool rows average the no-shock path without rebate and the price-plus-rebate package within each world. These χ^2 Wald sensitivities are approximate and potentially liberal with only 16–18 complete worlds per case; their calibration for this bank is unknown.

Effect	Y	Path	Estimate	W	df	p raw	p Holm	Conditional result	108-world bounds
TECH	B	drop	167/3456	1.38	1	.239	1	no rejection	[1/216, 71/216]
TECH	B	high	5/192	4.77	1	.029	.348	no rejection	[−1/48, 131/432]
TECH	B	pool	257/6912	2.72	1	.099	1	no rejection	[−7/864, 91/288]
TECH	J_b	drop	5/54	1.89	1	.169	1	no rejection	[5/54, 1/9]
TECH	J_b	high	41/432	5.41	1	.020	.280	no rejection	[1/12, 1/9]
TECH	J_b	pool	3/32	5.38	1	.020	.280	no rejection	[19/216, 1/9]
PROP	B	drop	281/2304, 425/2304	30.23	2	2.7×10^{-7}	4.6×10^{-6}	conditional rejection	[−13/36, 1/8]; [−43/144, 3/16]
PROP	B	high	1/384, 31/1152	2.60	2	.273	1	no rejection	[−139/288, 1/288]; [−11/24, 1/36]
PROP	B	pool	287/4608, 487/4608	31.04	2	1.8×10^{-7}	3.3×10^{-6}	conditional rejection	[−27/64, 37/576]; [−109/288, 31/288]
PROP	J_b	drop	25/144, 37/144	10.79	2	.005	.068	no rejection	[11/72, 13/72]; [17/72, 19/72]
PROP	J_b	high	1/96, 19/288	1.51	2	.470	1	no rejection	[−1/36, 1/72]; [1/36, 5/72]
PROP	J_b	pool	53/576, 31/192	11.02	2	.004	.065	no rejection	[1/16, 7/72]; [19/144, 1/6]
INT	B	drop	89/1152, 35/384	1.57	2	.456	1	no rejection	[−55/72, 5/24]; [−3/4, 2/9]
INT	B	high	19/576, 23/576	2.92	2	.232	1	no rejection	[−115/144, 25/144]; [−19/24, 13/72]
INT	B	pool	127/2304, 151/2304	2.62	2	.270	1	no rejection	[−25/32, 55/288]; [−37/48, 29/144]
INT	J_b	drop	1/6, 1/9	1.26	2	.532	1	no rejection	[1/9, 1/6]; [1/18, 1/9]
INT	J_b	high	19/144, 19/144	2.84	2	.242	1	no rejection	[1/12, 1/6]; [1/12, 1/6]
INT	J_b	pool	43/288, 35/288	2.96	2	.227	1	no rejection	[7/72, 1/6]; [5/72, 5/36]

Propensity. Under the independent-world sensitivity assumptions, using the 106 complete worlds, the λ categories differ in exploration quantity on the drop path (W = 30.2, 2 df, raw p = 2.7×10^{-7}, Holm p = 4.6×10^{-6}) and in the path average (W = 31.0, raw p = 1.8×10^{-7}, Holm p = 3.3×10^{-6}). The test does not reject on the high path (W = 2.60, raw p = .27, Holm p = 1). The 108-world bounds for both components of each rejecting contrast include zero: on the drop path, −13/36 to 1/8 and −43/144 to 3/16; in the path average, −27/64 to 37/576 (λ = .5 minus λ = 0) and −109/288 to 31/288 (λ = .95 minus λ = 0). The bank itself therefore does not sign these contrasts, because two record 1, λ = 0 worlds are partly unobserved and could have explored heavily in the missing rounds. In the path average, the one contrast that does not involve λ = 0, λ = .95 minus λ = .5, is point-identified at 25/576. For any exploration, the propensity test does not reject on any path (Holm p = .068 on the drop path and .065 in the path average). Those are non-rejections, although the 108-world bounds of both components exclude zero on the drop path and in the path average.

Technology. No technology test rejects, on either outcome or any path (Holm p from .28 to 1; Table S-F2). In the bank, however, missing data cannot reverse the observed sign of several technology differences: S minus I in exploration quantity on the drop path (1/216 to 71/216; worlds 36 positive, 44 zero, 26 negative, 2 ambiguous), and S minus I in any exploration on the drop path (5/54 to 1/9; 31, 54, 21, 2), on the high path (1/12 to 1/9; 15, 86, 5, 2) and in the path average (19/216 to 1/9). The quantity contrast on the high path and in the path average is not signed. The simple effects explain why the averages are not decisive:

in exploration quantity, the S−I difference is about zero at λ = 0 and positive at λ = .5 and .95 (drop path: −1/128, 5/72, 1/12; high path: 1/576, 5/144, 1/24). In the bank, S−I is non-negative in every identified cell with λ > 0. The conditional family does not reject a technology effect, and no mechanism is identified. Any S−I difference describes the implemented bundle, including the separated adviser's knowledge of λ.

Interaction. No interaction test rejects (Holm p = 1 in all six). The interaction compares each positive-λ S−I difference with the λ = 0 S−I difference. These summaries reuse the same six cells and are not independent replications. The λ = 0 cells depart from the zero benchmark; nevertheless, the J_b drop-path λ = 0 design difference is exactly zero, so its interaction components equal its positive-λ simple effects (1/6 and 1/9). For any exploration, all six interaction components have 108-world bounds that exclude zero, so in the bank the S−I gap in participation is larger at positive λ than at λ = 0; the conditional test does not reject. An additive interaction on a bounded share does not identify complementarity or a mechanism.

Both price paths. No effect rejects on both paths. The drop-path propensity rejection sits beside a high-path non-rejection, but a rejection on one path and a non-rejection on the other is not a test of whether λ changes the response to the package. The family contains no such test. In the complete-vector point estimates, most of the drop-path λ gradient coincides with λ differences in the drop-minus-high response: γ is (281/2304, 425/2304) on the drop path against (1/384, 31/1152) on the high path. The family does not test that difference, and any such difference would compare responses that start from λ-specific round-5 states.

S-F.3 Responses to the price-plus-rebate package

The factorial tests are not tests of the formal model, and against the model's own predictions the evidence is mixed. Assistance rose under the price-plus-rebate package in both designs, which is the sign Proposition 3 predicts. The drop-minus-high difference in assistance quantity is positive in 84 of 108 worlds under S (12 negative, 10 zero, 2 sign-ambiguous) and in 89 of 108 under I (4 negative, 14 zero, 1 sign-ambiguous). Its 108-world bounds exclude zero in both designs (S: 361/432 to 541/432; I: 245/216 to 65/54). No member of the test family covers assistance, and we claim sign consistency only, not magnitude. Case-level exploration responses are close to Proposition 4's null in record 0 at λ = 0, with 14 of 18 worlds at zero in each design (3 positive and 1 negative under S, 2 and 2 under I). They depart from it in record 1 at λ = .5, where 11 of 18 worlds under S and 5 of 18 under I explored more on the drop branch than on its matched high branch, and none explored less. The record 1, λ = 0 case is uninformative because two of its worlds are partly unobserved. The model derives no ordering between λ = .5 and λ = .95. Table S-F3 gives every case.

Table S-F3. Drop-minus-high response in exploration quantity B by case (108-world values; worlds + / 0 / − / ambiguous)

Case	D_S	Worlds (S)	D_I	Worlds (I)	$T = D_S - D_I$
rec0, λ = 0	1/36	3/14/1/0	0	2/14/2/0	1/36
rec0, λ = .5	11/72	7/10/1/0	1/6	8/10/0/0	−1/72
rec0, λ = .95	17/72	9/7/2/0	5/36	6/12/0/0	7/72
rec1, λ = 0	[−19/12, 7/4]	5/10/1/2	[−1/6, 7/18]	6/11/0/1	[−71/36, 23/12]
rec1, λ = .5	17/72	11/7/0/0	11/72	5/13/0/0	1/12
rec1, λ = .95	17/72	10/6/2/0	1/4	10/7/1/0	−1/72

Exploration responded less than assistance. Pooled over cases, the drop-minus-high difference in exploration quantity is bounded at −25/216 to 95/216 under S (45 worlds positive, 54 zero, 7 negative, 2 ambiguous) and at 13/144 to 79/432 under I (37, 67, 3, 1). The design difference in that response is not signed (−43/144 to 151/432).

S-F.4 Earnings, participation and retained capability

The package changed what workers bought and received more than what they could do. Consumption in drop branches is roughly 20 to 23.5 credits against 0.5 to 1.9 in high branches, but most of it is the rebate plus the fee saving; it is neither earnings nor a return on capability. Gross receipts over rounds 6–30 range from about 0.5 to 5.1 credits per branch, with drop-minus-high differences of about 0.3 to 3.6. (These scale

figures are Esther's arithmetic over complete worlds, not adopted statistics.)

The capability contrasts are small. Pooled over prices, the S–I difference in the change of retained performance is bounded at −7/4320 to 113/4320, with world signs split 47 positive, 24 zero, 35 negative and 2 ambiguous. Domains gained are rare: the pooled S–I difference in gains is bounded at −1/36 to 1/18, and 98 of 108 worlds show no S–I difference. The pooled S–I difference in capability losses is small, with 108-world bounds of −13/216 to −1/216 domains. It splits almost evenly across worlds (25 positive, 27 negative, 54 zero, 2 sign-ambiguous), and it reverses sign between endowments: +1/12 in record 0 against −11/54 to −5/54 in record 1. Loss can occur only in a held domain, and record 1 starts with two held domains against one in record 0, so exposure to loss differs mechanically by endowment. That difference in exposure is an alternative account of the reversal, not an established mechanism. Within each design, the drop branch lost fewer domains than its high branch under S (−5/36 to −7/108) and no more under I (−1/27 to 0), and its change in retained performance was higher (S: 23/1440 to 229/4320; I: 31/4320 to 37/1440). Design and package contrasts on capability are associations, not exploration effects, because capability can also form through paid work on adjacent-input tasks and through the helper's cards.

S-F.5 Robustness of the earlier analyses

Three post-collection masks treat observations possibly affected by transport recoveries, retries and redraws as unknowns while keeping all 108 worlds. No mask reverses the sign of any identified interval in the original family; 10, 13 and 23 of the 62 nonzero identified intervals come to include zero under the three classes. The dispatch-time cut, which masks positions 99–108, leaves 61 of 62 including zero. That shows lost identification, not contrary evidence, and the cut is not a counterfactual for the original stop policy. Leaving out one world at a time changes the sign of none of the nine members with raw $p < .05$; the largest receipt contrast moves by about two-fifths when one world is omitted. For the two-factor analysis, the repetitions on expression-complete cohorts, by endowment and with each world left out (17,172 deletion records) add no tests; the cut diagnostics recompute effects and standard errors without p-values. No member of the 18-test family has a singular contrast covariance, but 6 of the 36 primary case groups have zero estimated variance in the I design, all on the high path. That is a descriptive degeneracy, not certainty. Sections S-F.6 to S-F.8 list every factorial cell, simple effect and headline display.

S-F.6 All 36 cells

Primary 108-world bounds and 106-world complete-vector estimates with standard errors and unadjusted marginal normal 95% intervals. Pool rows average, within each world, the no-shock path without rebate and the price-plus-rebate package; they are not a third experiment.

Table S-F4. All 36 cells (displays159.csv; factorial-cells-complete36.csv)

Cell	108-world bounds	Complete-vector estimate	SE	Marginal normal 95%
FC-B-drop-S-0	[11/144, 131/144]	29/384	0.0227	[0.031, 0.120]
FC-B-drop-S-05	17/72	17/72	0.0458	[0.146, 0.326]
FC-B-drop-S-095	11/36	11/36	0.0622	[0.184, 0.427]
FC-B-drop-I-0	[11/144, 31/144]	1/12	0.0255	[0.033, 0.133]
FC-B-drop-I-05	1/6	1/6	0.0525	[0.064, 0.270]
FC-B-drop-I-095	2/9	2/9	0.0503	[0.124, 0.321]
FC-B-high-S-0	[1/48, 41/48]	13/576	0.0127	[−0.002, 0.048]
FC-B-high-S-05	1/24	1/24	0.0153	[0.012, 0.072]
FC-B-high-S-095	5/72	5/72	0.0235	[0.023, 0.115]
FC-B-high-I-0	[1/48, 23/144]	1/48	0.0113	[−0.001, 0.043]
FC-B-high-I-05	1/144	1/144	0.0069	[−0.007, 0.021]
FC-B-high-I-095	1/36	1/36	0.0162	[−0.004, 0.059]
FC-B-pool-S-0	[7/144, 127/144]	113/2304	0.0130	[0.024, 0.075]
FC-B-pool-S-05	5/36	5/36	0.0258	[0.088, 0.189]
FC-B-pool-S-095	3/16	3/16	0.0337	[0.121, 0.254]
FC-B-pool-I-0	[7/144, 3/16]	5/96	0.0144	[0.024, 0.080]
FC-B-pool-I-05	25/288	25/288	0.0289	[0.030, 0.144]
FC-B-pool-I-095	1/8	1/8	0.0269	[0.072, 0.178]
FC-Jb-drop-S-0	[5/18, 11/36]	13/48	0.0771	[0.120, 0.422]

Cell	108-world bounds	Complete-vector estimate	SE	Marginal normal 95%
FC-Jb-drop-S-05	19/36	19/36	0.0822	[0.367, 0.689]
FC-Jb-drop-S-095	7/12	7/12	0.0844	[0.418, 0.749]
FC-Jb-drop-I-0	[1/4, 5/18]	13/48	0.0771	[0.120, 0.422]
FC-Jb-drop-I-05	13/36	13/36	0.0811	[0.202, 0.520]
FC-Jb-drop-I-095	17/36	17/36	0.0822	[0.311, 0.633]
FC-Jb-high-S-0	[1/12, 5/36]	13/144	0.0509	[−0.010, 0.190]
FC-Jb-high-S-05	1/6	1/6	0.0610	[0.047, 0.286]
FC-Jb-high-S-095	2/9	2/9	0.0707	[0.084, 0.361]
FC-Jb-high-I-0	[1/12, 1/9]	1/12	0.0452	[−0.005, 0.172]
FC-Jb-high-I-05	1/36	1/36	0.0278	[−0.027, 0.082]
FC-Jb-high-I-095	1/12	1/12	0.0452	[−0.005, 0.172]
FC-Jb-pool-S-0	[13/72, 2/9]	13/72	0.0469	[0.089, 0.272]
FC-Jb-pool-S-05	25/72	25/72	0.0561	[0.237, 0.457]
FC-Jb-pool-S-095	29/72	29/72	0.0523	[0.300, 0.505]
FC-Jb-pool-I-0	[1/6, 7/36]	17/96	0.0469	[0.085, 0.269]
FC-Jb-pool-I-05	7/36	7/36	0.0462	[0.104, 0.285]
FC-Jb-pool-I-095	5/18	5/18	0.0476	[0.184, 0.371]

S-F.7 Simple effects of technology by λ

The S−I difference in each λ category (complete-vector estimates). The interaction components are differences of these values; they are not independent replications.

Table S-F5. S−I simple effects δ by λ and the technology main effect τ (complete-vector estimates; factorial-scalar-intervals.csv)

Y	Path	δ(λ = 0)	δ(λ = .5)	δ(λ = .95)	τ
B	drop	−1/128	5/72	1/12	167/3456
B	high	1/576	5/144	1/24	5/192
B	pool	−7/2304	5/96	1/16	257/6912
J_b	drop	0	1/6	1/9	5/54
J_b	high	1/144	5/36	5/36	41/432
J_b	pool	1/288	11/72	1/8	3/32

S-F.8 Headline displays with world distributions

Primary 108-world bounds and the distribution of world-level values (positive, zero, negative, sign-ambiguous). Between-λ contrasts are not paired across worlds. Display IDs follow the frozen specification.

Table S-F6. Headline displays (displays159.csv; world-spreads.csv summed over cases)

ID	Outcome	Definition	108-world bounds	Worlds + / 0 / − / ambiguous
A01	B	L_drop	[1/216, 71/216]	36 / 44 / 26 / 2
A02	B	L_high	[−1/48, 131/432]	16 / 85 / 5 / 2
A03	B	(L_drop+L_high)/2	[−7/864, 91/288]	39 / 42 / 25 / 2
A04	Jb	L_drop	[5/54, 1/9]	31 / 54 / 21 / 2
A05	Jb	L_high	[1/12, 1/9]	15 / 86 / 5 / 2
A06	Jb	(L_drop+L_high)/2	[19/216, 1/9]	34 / 53 / 19 / 2
B01	B	lambda 0 level M(l;Y)	[7/144, 77/144]	between-λ; not paired
B02	B	lambda 0.5 level M(l;Y)	65/576	between-λ; not paired
B03	B	lambda 0.95 level M(l;Y)	5/32	between-λ; not paired
B04	B	M(lambda 0.5)-M(lambda 0)	[−27/64, 37/576]	between-λ; not paired
B05	B	M(lambda 0.95)-M(lambda 0)	[−109/288, 31/288]	between-λ; not paired
B06	B	M(lambda 0.95)-M(lambda 0.5)	25/576	between-λ; not paired
B08	Jb	lambda 0 level M(l;Y)	[25/144, 5/24]	between-λ; not paired
B09	Jb	lambda 0.5 level M(l;Y)	13/48	between-λ; not paired
B10	Jb	lambda 0.95 level M(l;Y)	49/144	between-λ; not paired

ID	Outcome	Definition	108-world bounds	Worlds + / 0 / – / ambiguous
B11	Jb	M(lambda 0.5)-M(lambda 0)	[1/16, 7/72]	between-λ; not paired
B12	Jb	M(lambda 0.95)-M(lambda 0)	[19/144, 1/6]	between-λ; not paired
B13	Jb	M(lambda 0.95)-M(lambda 0.5)	5/72	between-λ; not paired
D01	B	D_S	[−25/216, 95/216]	45 / 54 / 7 / 2
D02	B	D_I	[13/144, 79/432]	37 / 67 / 3 / 1
D03	B	T = D_S-D_I	[−43/144, 151/432]	36 / 44 / 26 / 2
D04	Aq	D_S	[361/432, 541/432]	84 / 10 / 12 / 2
D05	Aq	D_I	[245/216, 65/54]	89 / 14 / 4 / 1
D06	Aq	T = D_S-D_I	[−53/144, 17/144]	42 / 8 / 56 / 2
D07	Jb	D_S	[31/108, 17/54]	39 / 60 / 7 / 2
D08	Jb	D_I	[31/108, 11/36]	35 / 69 / 3 / 1
E01	Epost	(L_drop+L_high)/2	[719/5184, 6389/5184]	43 / 38 / 25 / 2
E02	Epost	D_S	[1867/10368, 21307/10368]	36 / 66 / 4 / 2
E03	Epost	D_I	[2105/3456, 3185/3456]	28 / 74 / 5 / 1
E06	F30	(L_drop+L_high)/2	[−1/24, 7/72]	19 / 71 / 16 / 2
E07	F30	D_S	[−1/36, 17/108]	14 / 85 / 7 / 2
E08	F30	D_I	[1/12, 19/108]	16 / 89 / 2 / 1
E11	dR	(L_drop+L_high)/2	[−7/4320, 113/4320]	47 / 24 / 35 / 2
E12	dR	D_S	[23/1440, 229/4320]	39 / 54 / 13 / 2
E13	dR	D_I	[31/4320, 37/1440]	34 / 54 / 19 / 1
E21	G	(L_drop+L_high)/2	[−1/36, 1/18]	4 / 98 / 4 / 2
E26	Loss	(L_drop+L_high)/2	[−13/216, −1/216]	25 / 54 / 27 / 2
E27	Loss	D_S	[−5/36, −7/108]	10 / 77 / 19 / 2
E28	Loss	D_I	[−1/27, 0]	15 / 73 / 19 / 1
H201	R30-R0	S/drop mean change, rec0 one-card lambda .95	−7/240	3 / 11 / 4 / 0
H202	R30-R0	I/drop mean change, rec0 one-card lambda .95	−11/240	1 / 11 / 6 / 0
H203	H30-H5	S/drop mean change, rec0 one-card lambda .95	−13/720	4 / 9 / 5 / 0
H204	H30-H5	I/drop mean change, rec0 one-card lambda .95	−1/80	4 / 10 / 4 / 0

S-F.9 Reviewer positions on the disclosure for the conditional tests

The paper uses Ada's wording (Section S-F.1). Toby's proposed amendment, which he does not press as blocking, reads: "These χ^2 Wald sensitivities are approximate. Even under their own independent-world assumptions, a χ^2 reference that treats a covariance estimated from 16–18 complete worlds per case as known tends to reject too often. For this heterogeneous, potentially dependent bank, the actual false-positive rate and the direction and size of any further error are uncalibrated. Holm adjustment does not repair invalid marginal p-values or post-outcome analytical selection. Marginal normal intervals are unadjusted displays and create no additional rejection or equality claims." Neither text changes which members reject.

S-G. Formal model in full (W2)

Preserved from the 24 October draft (SHA256 `0b43d62a…`), with its results unchanged. Appendix, table and figure letters have been renumbered for this paper (old A→D, B→E, C→F, D→G, E→H, F→I); section numbers 6.x–9.x cited inside refer to the 24 October draft (`ICIS Full Draft for Review 2026-10-24.md`).

Provenance. The base is the model of 17 October (`outputs/AI Design and Worker Capacity - Revised Model and Hypotheses 2026-10-17.md`, sections M1–M6 and A1–A6), drafted by agents in the room's review graph from the author's review comments; the author confirmed the four-branch structure. It was updated by the hidden-domain specifications HD5-1 to HD5-4 (Athena, 18–19 October) and the checks of Esther and Toby. Howard assembled this statement on 19 October. Section references in brackets give the source.

W2.1 Primitives

Rounds t = 1, …, 30, with time T = 1 per round. Five domains d ∈ {d0, …, d4}; d0 and d2 are easy, d1 and d3 are step, d4 is two-step. They are shown to agents as Service A to E in a seeded order. The initial record holds the true card for d0 (record 0) or for d0 and d1 (record 1); L_0 is the set of domains absent at the start. λ ∈ {0, .5, .95}. Prices: p_t = 1 for t ≤ 5; p_t ∈ {.25, 1} for t ≥ 6 by branch [M6].

W2.2 Mechanisms and the public grammar

Inputs u ∈ {0, …, 1023}, v ∈ {0, …, 3}, c ∈ {A, B}, with z = 1[c = B]. The mechanism is f(u, v, c) = k + αu + βv + γz + δg(u) + ηuv + ζzv. Affine: g = 0 and δ = η = ζ = 0. Step: g(u) = 1[u ≥ τ], τ ∈ {256, 768}. Two-step: g(u) = 1[u ≥ τ] + 1[u ≥ τ + 256], τ ∈ {256, 512}. Supports: k ∈ 100…999; α, β ∈ 1…9; γ ∈ ±(1…19); hard δ ∈ 5…31, η ∈ 1…3, ζ ∈ 1…9 [HD5-4 §2]. Every model payload carries this grammar verbatim, ending "Some domains may share constants; no pair or shared value is supplied here." A wrong form or threshold loses at least a quarter of uniform-input accuracy in the worst case [HD5-4 §3].

Domains share constants along a directed Hamiltonian cycle drawn per world before any parameter. Edge slots are α, β, α, β, γ in order, and the predecessor h(d) supplies d's adjacent inputs. In each domain, paid slots 1 and 5 of 8 are adjacent-input tasks. The task gives four rows (u_j, v_j, c_j, y_j) of h(d) and asks for f_d(u, *v, c), where u* = (Σ y_j) mod 1024. No map, edge or tag appears in any payload [HD5-2 §5; STUDY-CEILING-3000 Addendum 1, MF-S6].

W2.3 Choices and feasibility

z = (x, a, b) on {0, .25, .5, .75, 1}^{3}, with a ≤ x. Each record update costs τ_M = .25 of the round. Let i_P = 1[x > 0] for the production update and r = 1 − x − .25i_P. Then i_W = 1 if b > 0 or r > 0, and e = 1 − x − .25(i_P + i_W) ≥ 0 is unpaid workroom time. b ∈ [0, 1] is independent of e. Purchases must satisfy p(a + b) ≤ w before delivery. There are 34 feasible triples, and x = 1 is infeasible [HD5-2 §3]. The worker routes paid attempts across domains and names one target domain for b, which must be absent from the initial record [HD5-1 §4; HD5-2 §3].

W2.4 Production, pay and cash

With m = 4x baseline and A_h = 4a helper attempts, ρ = 1 − a/(2x). Q_d = (C^base_d + ρC^help_d)/4, and when all attempts succeed Q = x + a − a^2/(2x) exactly [HD5-2 §2]. Pay per baseline-equivalent answer, before the common scale s: d0 1/2; d1 3/4; d2 1 (even world) or 1/8 (odd); d3 1/8 or 5/4; d4 3/2 or 1/8 [HD5-1 §4]. The scale was frozen at s = 1/2 by the ordered rule "1/2, then 1/4" (Esther's condition 1; Avery study build v2). Y = Σ_d sπ_d(C^base_d + ρC^help_d). Cash follows M3: w_1 = 2, reserve E_1 = 30, fee F = p, rebate 1 − p, S = w − p(a + b) + Y + (1 − p), c = S/2 for t < 30 and c = S at t = 30. Conservation c + w' + E' + p(a + b) + F = w + E + Y holds every round [M3, A2]. The drop branches receive a direct rebate of .75 × 25 = 18.75.

W2.5 Exploration and acquisition

A purchase of B = 4b quarters at dose d = 2 delivers 2B new rows of the target and one analysis call. The first five rows of each domain are P1 = (0,0,A), P2 = (1,0,A), P3 = (0,1,A), P4 = (0,0,B) and P5 = (1023,3,B).

They identify an affine mechanism exactly and detect a hard one; at d = 2 this needs b = .75. Later rows are uniform and truth-blind [HD5-3 §2; HD5-4 §4]. The analysis sees at most 16 rows per call, the earliest four and the newest [HD5-2 §5]. The candidate card enters the record only through the worker's own update, and the domain counts as acquired only when its card is applied to fresh tasks. A latent-domain helper buys its own rows inside the price of a; its inference never enters the worker's record [HD5-1 §4].

Benchmark acquisition law (surrogate, for the witness only; not measured competence). Easy: acquired with probability e_0 = 1/2 once n ≥ 5 rows have been delivered. Hard: probability h·max(0, n − 4)/(4d − 4), with h = 3/8 at the full purchase (n = 8). Adjacent-input encounters: 1/8 for an easy upstream domain, 0 for a hard one (1/16 in sensitivity). Known-domain helper accuracy is .9, no-card helper accuracy .25, own answers in a held domain 1, and own answers in a latent domain 0 [HD5-2 §6; HD5-4 §8]. Esther's panel, written before the hard-family result, shows the exploration-margin price response surviving at s ≤ 1/2 for full-budget hard acquisition up to 3/4 (Esther dose check §1; DEC-…-OPTION-2-HARD-CRITERION item 2). The calibrated hard band of 4 to 12 successes in 16 corresponds to rates from 1/4 to 3/4. Its ceiling derives from the Q4 failure clause, not from an economic ceiling (Toby MF-H1).

W2.6 Record and retention

The record R is at most five cards {domain, created_round, form, params, procedure, error_cue}. Numbers are allowed only in typed parameters; task identifiers, input–output lookups and answers are banned; an invalid update leaves R unchanged [HD5-1 §5]. There are two updates per round. After paid work, the update reads the round's tasks, including adjacent-input rows, own answers and their correctness, and known-helper answers with their method cards. After the workroom, it reads the workroom attempts and the purchased evidence. Both opportunities are identical across designs. Unlike M4 of 17 October, paid work can therefore update the record.

W2.7 Advice

The separated adviser receives λ, the objective, a two-row numeric ledger, R and the shared state. The integrated adviser receives only the shared state: round, horizon, price, cash, aggregate certification, rules and catalogue. Each returns at most three feasible plans labelled current_receipts, continuation_sensitive or conditional. The integrated design replays its current-round advice exchange into the helper's final call. Both cost one reserve credit [M5; HD5-1 §7].

W2.8 Estimands

D_S(M) = M_{S,drop} − M_{S,high}, D_I(M) = M_{I,drop} − M_{I,high} and Θ(M) = D_S(M) − D_I(M), per world and averaged within each of the six initial cases with equal weights [M6; Ada SAP §3]. Outcomes [Ada SAP §2]:

- F_30: domains served with a held card in round 30.
- E_ext: receipts from L_0.
- H_30: equal-weight certification over the five domains.
- R_30 and U_30: retained and scrubbed evaluation accuracy.
- H_5: certification at the shock, for H_30 − H_5.
- K_dt (companion): a valid complete card with R_dt ≥ 6/8 and R_dt − U_dt ≥ 2/8.

W2.9 Propositions, proofs in brief

The statements as written on 19 October. Until 26 October the four statements below stood in Section 5 of the main text. They were moved here unchanged when the main text was reorganized around the research questions; Section 5 now gives only the short benchmark explanations needed to read the experiment. The derivations follow the statements.

Proposition 1 (own-record benchmark). If both advisers convey only functions of the worker's own record and add no payoff information, an exact optimiser under her beliefs cannot gain from either. With common tie-breaking, both designs choose identically. A design effect therefore requires bounded processing or another friction, or a channel through the helper's context.

Proposition 2 (myopic benchmark). At $\lambda = 0$, or in round 30, an exact maximiser chooses $b = 0$ and the largest feasible x. Exploration pays only in later rounds, and output rises in x.

Proposition 3 (assistance responds to price). With helper accuracy q and pay π in a held domain, the continuous optimum at fixed x, ignoring liquidity, is $a^*/x = 1 - p/(4\pi q)$, clipped to $[0, 1]$. A lower price raises assistance. In the exact benchmark, the assistance path differs between the price branches in every cell.

Proposition 4 (exploration responds to price in some cells only). Under the benchmark acquisition law of Appendix W2 and the frozen pay, the post-shock exploration path of an exact maximiser differs between the drop and high branches in some cells and is identical in others. It differs with one initial card at $\lambda = .5$. It is identical with two initial cards at $\lambda = .5$ and at $\lambda = 0$. At $\lambda = .95$ the relaxed benchmark almost never changes its exploration path; the cash-constrained solve for these cells did not complete. The design therefore does not force a price response on the exploration margin.

The interaction Θ is a reported estimand with no predicted sign. The model orders neither bounded design above the other (Propositions 1 and 2).

Derivations.

Proposition 1 restates P0 of 17 October [M6, A3]. If adviser inputs are functions of R and add no payoff information, an exact optimiser under her beliefs can reproduce either advice and ignore it. The proposition does not cover the helper-context channel.

Proposition 2 revises C2. At $\lambda = 0$ the objective is current c. A purchase $b > 0$ costs $p \cdot b$, yields nothing in the round and forces the workroom update (i_W = 1), which lowers the feasible x. With $b = 0$ and $x > 0$, $Q_x = 1 + a^2/(2x^2) > 0$, so $x = .75$, the largest feasible value. The corner moves from (1, a, 0) in the 17 October model to (.75, a, 0). The witness confirms it: no $\lambda = 0$ cell explores.

Proposition 3 is the help margin. At fixed x, with known helper accuracy q and pay π, expected helper receipts are $4\pi q(a - a^2/(2x))$. Ignoring liquidity and future returns, the optimum is $a^*/x = \text{clip}(1 - p/(4\pi q), 0, 1)$ [HD5-2 §2]. It replaces $a/x = 1 - p/r(h)$ [M2, A3]. In the exact witness, the a path changes between the price branches in 24/24 cycles in every cell at both scales.

Proposition 4 is the exploration margin, from the exact witness [DEC-…-ECONOMY 23:55 addendum; STUDY-CEILING-3000 Addendum 2; Avery study build v2]. The optimiser knows the full hidden cycle, so it is an upper-information benchmark. It optimises rounds 1–5 under a persistent price of 1, then forks the round-5 state and re-optimises at the observed price. Entries in Table W2.1 give P(post-shock (b, b_domain) path differs), averaged over 24 anchored cycles. NA means the cash-constrained solve was not completed (2,000,001 evaluations exhausted), never zero.

Table W2.1. Exact witness, P(exploration path differs between drop and high), at the frozen scale s = 1/2

Record	λ	Even world, relaxed	Even world, constrained	Odd world, relaxed	Odd world, constrained
0	0	0	0	0	0
0	.5	49575/65536	49575/65536	1	1
0	.95	$\approx 4.8 \times 10^{-6}$	NA	$\approx 3.1 \times 10^{-6}$	NA
1	0	0	0	0	0
1	.5	0	0	0	0
1	.95	$\approx 4.1 \times 10^{-6}$	NA	$\approx 7.9 \times 10^{-6}$	NA

At $s = 1/4$, only record 0, $\lambda = .5$, even world diverges (1/4 in both solves). Both scales met the criterion, and the ordered rule selected 1/2. Sensitivity coverage is partial: of the 720 core configurations, 357 were solved exactly, relaxed, and 363 were resource-limited. Two learned-on-encounter-map variants are unimplemented. Esther's earlier approximate analysis placed the $\lambda = .95$ cells at the b ceiling before the shock (Esther economy check, Summary). That is approximate and is not witness evidence.

Contract check (C1 of 17 October). Equal full inputs, session state and response randomness imply equal actions. No deterministic equality is inferred from an API seed [M6, A3].

C3 of 17 October (evidence sensitivity) rests on a latent responsiveness θ with a posterior. The hidden-domain model has no θ; it uses fixed acquisition rates. C3 is retained only as a statement about the analytical special case of 17 October and is not tested.

W2.10 Hypotheses

H1 (production): $D_S(a) > 0$ and $D_I(a) > 0$. This is new as a hypothesis; Proposition 3 and M2 give it as a property of the benchmark. H2 (capability) is C4 of 17 October with its focal case moved from the empty record to record 0 (one card), $\lambda = .95$, drop branch: $R_30 - R_0 > 0$ and $H_30 - H_5 > 0$, separately for S and I. Θ has no predicted sign. The hypotheses of 16 October (a "slide" and a gated "breakout") are superseded.

W2.11 Model changes since 17 October

Table W2.2. Changes to the model and instrument, 17–19 October 2026

#	What changed	Authorising record	Authorised by	Kind
1	Sample 12 → 192 worlds, 32 per case	SAP v3 (17 Oct); ROOM.md 17 Oct 15:26	Author ("300 is fine.")	Design
2	Within-design round-5 fork adopted (the 17 Oct "optional refinement")	SAP v3 (17 Oct)	Ada, in SAP v3; no separate approving record located	Design
3	Outcome-conditioned score cutoffs removed; usable-assistance check; procedure-only memory kept	measurement-correction (18 Oct)	Author ("go for it.")	Instrument (subset task, superseded)
4	Response headroom; numerical fix; funding pause	response-headroom; funding-recovery (18 Oct)	Author (headroom: "we should budget more generous room"); numerical fix and funding pause: no authorising decision located (ledger imports 15 and 21)	Instrument
5	Task family: hidden-mechanism domains. b becomes dry exploration priced in compute; validation by paid tasks; certification tariff removed; capability = domains served	DEC-20261018-ICIS-HIDDEN-DOMAINS	Author	**Theory**
6	Charged observation service; memory validator admits typed constants; a record update after production in both arms	same, items 4–5 (Astra's corrections)	Author	**Theory** (retention channel)
7	Concave helper credit ρ; b unbound from time; $\tau_M = .25$; hidden map with adjacent-input tasks; symmetric production memory	DEC-…-HIDDEN-DOMAINS-ECONOMY	Author ("Authorize them.")	**Theory**

#	What changed	Authorising record	Authorised by	Kind
8	Witness criterion: post-shock (b, b_domain) path, not (x, a, b)	ECONOMY, 23:55 addendum	Leo, under repair authority	Benchmark
9	Repair-and-go rule	DEC-…-REPAIR-AND-GO	Author	Governance
10	Fifth identifying point; parsimony instruction; effort calibration	DEC-20261019-ICIS-Q4-IDENTIFICATION (+ Add. 1)	Leo, under repair authority, after a reached failure	Instrument
11	Grammar in every task payload; answer after summary, no JSON mode; thinking budget 2,048 → 4,096	DEC-…-ROUTE-1-ACTOR-REPAIR (+ Add. 1–2)	Author ("Authorized for 1."), after a reached failure	Instrument
12	Observation dose 2 per quarter, paid prefix; analysis effort frozen at low	DEC-…-OPTION-1-OBSERVATION-DOSE	Author ("go with option 1"), after a reached failure	Instrument; sets acquisition probabilities
13	Hard family: step and two-step with two-valued thresholds; hard success = full-function equivalence and accuracy ≥ .70; band 4–12	DEC-…-OPTION-2-HARD-CRITERION	Author ("go with option 2"), after a reached failure	Instrument; changes what counts as acquisition
14	Pay scale by ordered rule (1/2, then 1/4); frozen at 1/2	Esther dose check, condition 1; Avery study build v2	Seats, under the author's decisions	Economy
15	Ceiling $3,000; 48 workers; K_frozen from throughput	DEC-…-STUDY-CEILING-3000 (+ Add. 1)	Author; Toby MF-S1–S8	Operations
16	Grammar also reaches decision, adviser and adoption payloads	STUDY-CEILING-3000 item 4; Add. 1	Leo (provisional); Toby (RATIFY)	Instrument; estimand "given the grammar"
17	Witness pass judged on solved cells; time box	STUDY-CEILING-3000 Add. 2	Leo, under repair authority	Benchmark coverage
18	Compute stop Friday 23 Oct 18:00	DEC-20261019-ICIS-STUDY-STOP-FRIDAY	Author	Operations
19	Pause and failure semantics: one world's failure never halts the study; resume from the cursor; never re-freeze K	STUDY-CEILING-3000 Add. 3	Leo, under repair authority	Instrument (operations)
20	Live rulings: crash = halt and resume; on-disk trace spill with one controlled pause; a provider budget overrun scored as a wrong answer, world resumes	STUDY-CEILING-3000 Add. 4–6	Leo, under repair authority	Instrument (interface)
—	λ as a discount factor; half consumption; rebate 1 − p	M1, M3 of 17 Oct	(unchanged since 17 Oct)	—

Note on rows 7, 8 and the K rule. Rows 7 and 8 rest on the economy decision, whose addenda were written inside the file. The K rule of the repair-and-go decision (used in rows 1 and 15) was replaced in place on 18

October at 22:26 UTC (Table W1.1).

S-H. Experimental design and qualification record (W3)

Preserved from the 24 October draft (SHA256 `0b43d62a…`), with its results unchanged. Appendix, table and figure letters have been renumbered for this paper (old A→D, B→E, C→F, D→G, E→H, F→I); section numbers 6.x–9.x cited inside refer to the 24 October draft (`ICIS Full Draft for Review 2026-10-24.md`).

Provenance. Assembled by Howard on 19 October 2026 from the specifications (HD5-1 to HD5-4, Athena), the checks (Toby, Esther, Scully), Ada's analysis plan and Avery's run handoffs. The standalone methods section, revised the same day, is `outputs/Howard - ICIS Methods Section, Hidden-Mechanism Design 2026-10-19.md`.

W3.1 World, schedule and roles

Each world has five domains, 30 rounds and four branches. A world needs 110 unique round transitions (5 per design before the fork, then 25 per branch) and 1,640 unique assessment calls: common round-0 assessments, separate round-5 certification per design, certification at rounds 10–30 and paired evaluation at rounds 15 and 30. The total is 2,300 to 2,960 calls per world, depending on actions [Avery study build]. Within a round the order is adviser → decision → own paid answers → helper answers and adoption → production update → workroom and b service → workroom update → settlement → scheduled assessments [HD5-1 §4]. All production uses the record as it stood at the start of the round. The study bank holds 422,400 tasks, including 57,600 adjacent-input tasks, and 960 mechanisms disjoint from the 1,040 used in qualification [Avery study build v2].

W3.2 Qualification gates, full rules

Table W3.1. The six gates, frozen before any paid call

Gate	Pass rule	Attempt 5, development	Attempt 5, fresh
Q1 Oracle	64 worlds per band; one-sided 95% Clopper–Pearson lower bound ≥ .85 easy, ≥ .70 hard; each hard subtype point rate ≥ .70	—	Easy 63/64 (lower .928); hard 62/64 (lower .905); step 30/32, two-step 32/32
Q3 Thinking room	In the Q1 calls, ≤ 6/64 per band at 4,096 reasoning tokens; no unknown or over-cap counts	—	0/64 in each band
Q2 Empty ceiling	Same tasks, empty record; upper bound ≤ .40 per band	—	0/64 each (upper .046)
Q5 Helper	Luna with the true card, ≥ 58/64 per band	—	Easy 59/64; hard 60/64
Q4 Inferability	Easy: one purchase at b = .75 (6 rows), ≥ 18/24 cards equivalent to truth on all 8,192 inputs. Hard: b = 1 (8 rows); ≥ 4 successes and ≥ 4 failures in 16, 2–6 successes per eight-world subtype; success = complete, valid, no cap hit, equivalent and paid accuracy ≥ .70	Easy 22/24; hard 9 success / 7 failure (step 5/8, two-step 4/8)	Easy 23/24; hard 11/5 (step 5/8, two-step 6/8)
Q6 Acquisition	24 easy worlds; one worker update from an empty record using the Q4 artifact; four fresh tasks per world, with record vs empty; mean gain ≥ .30, ≥ 18 positive worlds, ≥ 18 valid complete cards	90/96 vs 0/96; gain .9375; 23 positive; 22 cards	91/96 vs 0/96; gain .948; 23 positive; 23 cards

Order: Q1 and Q3 (same calls) → Q2 → Q5 → verify the frozen dose → development Q4 → development Q6 → fresh Q4 → fresh Q6. The run stops at the first gate reached that fails. Development hard Q4 re-sends the calibration requests and is a consistency check only; fresh Q4 is the independent test (Toby MF-D3, MF-H4). Q6 certifies easy carry-and-apply only, not hard retained acquisition. Attempt 5 made 896 calls (688 Qwen, 208 Luna) in 47.5 minutes at 8 workers, for $0.289, with no retries or pauses. One invalid memory update was kept as a failure, without repair [Avery attempt 5].

W3.3 The rule for failures

An interface failure is repaired prospectively with a fixture, checked by Scully on the diff and rerun under a new number. Interface failures are those of output shape, prompt–validator mismatch, token-cap sizing, provider JSON abort, transport, throttle, ledger or launch wiring. A frozen gate that is reached and fails is a result: it stops admission, no response is regraded and no threshold changes after data. Every attempt is kept and disclosed [DEC-…-REPAIR-AND-GO items 1–2, 5].

W3.4 Paid runs

Table W3.2. Paid qualification runs, 18–19 October 2026

Run	Calls; cost	Outcome	What it established	What followed
1	1; $0.0004	Interface stop	The actor's routing list did not match its allocation	Format repair
2	64; $0.026	Failed development Q4: easy 7/24; hard 2/16 at ≥ .70	Four fixed observations could not identify the form	Amendment 1: a fifth identifying point; parsimony instruction
3, 3b	330; $0.136	3: interface overflow. 3b: passed development Q4 (23/24; 8/16); failed development Q6, 1/96 vs 0/96	22 of 24 inferred cards were correct, yet answers failed: an answer field that disagreed with the actor's reasoning, no grammar in the task, a binding budget	Repair (the author's route 1): grammar in every payload; answer after summary, no JSON mode; budget 2,048 → 4,096
4	432; $0.142	Q1, Q2, Q3, Q5 passed; failed the analysis-effort calibration (low 13/3, medium 15/1, high 11 with 5 cap hits)	With 20 observations hard inference rarely failed	Amendment 2: fewer observations
Dose calibration	48; $0.028	No eligible dose: 9, 11, 13 of 16 at 8, 12, 16 rows (band 4–8)	A wrong step card could serve the old one-point window form	Amendment 3: two-step form; two-valued thresholds; equivalence required
5	896; $0.289	All six gates passed	The instrument is certified	Study build

Note. Also paid: the HD5-4 calibration (48 calls, $0.028698; 9/16, 10/16, 10/16 equivalent hard cards at doses 2, 3, 4; dose 2 frozen) and three channel-fixture units for the actor repair (32 + 28 + 13 calls; $0.028538). The C1 candidate was eliminated at the interface stage; C2 passed fidelity 16/16 and 24/24 interface checks. In total, qualification made 1,892 calls for $0.679. Conservative liability recorded before attempt 5 across all earlier paid work was $15.43. That is a reservation bound, not an invoice.

W3.5 The amendments and their safeguards

Three design amendments followed reached failures, each before any treatment data and none selected on a treatment contrast [DEC-…-OPTION-2-HARD-CRITERION item 7]. Each was specified by Athena, tested by Toby for tuning, leakage and failability, and checked by Esther on the economics, before Avery's build and Scully's recheck. Toby's MF-H1, adopted verbatim, governs the hardest case:

> "The 12/16 ceiling and the 6/8 subtype ceiling derive from the Q4 hard two-sided clause (≥4/16 below .70 or invalid; T1 M2, 18 October; DEC-20261019-ICIS-Q4-IDENTIFICATION item 3), and the 4/16 floor from DEC-20261019-ICIS-OPTION-1-OBSERVATION-DOSE item 1; Esther's dose-check §1 panel shows only that [1/4, 3/4] is economically admissible, not that 3/4 is a ceiling. The paper states that this band, applied to the old family and old criterion, would have admitted dose 2 (9/16; step 4/8, window 5/8) and dose 3 (11/16), and that it is admissible only jointly with the equivalence criterion, under which the failed calibration would score 0/16 at every dose."

Esther reported she "would have written the same band had the counts been 5/7/9." The actor repair after attempt 3b also followed a reached failure. Toby first showed that the channel fault alone did not explain it (recomputed gain .302, 12/24 positive worlds), and the classification was then reserved to the author, who called the missing grammar a prompt defect. The same repair raised the thinking budget.

W3.6 Sample, stopping and money

The study uses 192 worlds in a frozen order. Each block of 12 holds each initial case twice, once per pay parity, and the order was drawn by a rule written before any parameter or witness cell was seen (MF-S7). It runs 48 workers in parallel. K_frozen is computed by a script whose only inputs are pooled transport timestamps over exactly the first 60 minutes after the first provider response and a per-world call constant of 2,960 fixed before launch (MF-S1, MF-S2). It is the largest multiple of 12 ≤ 192 with projected duration × 1.20 ≤ the time remaining to Friday 23 October 18:00 London [DEC-20261019-ICIS-STUDY-STOP-FRIDAY]. The analysed set is the first K_frozen positions. Incomplete work is NA within bounds, and complete-case analysis is a sensitivity only (MF-S3). From the freeze, analysed worlds have dispatch priority (MF-S4). The cut time is computed once (MF-S5). Below K = 96 the analysis is descriptive, and stopping is the author's decision. The ceiling is $3,000 under a conservative reservation of $2,899.88 for the full design; the realised cost is not reported in the outputs used for this draft. The funding pause pauses dispatch and never cancels. The study launched at 13:14:20 London on 19 October (run `run_28f07a321e5f`, manifest `962d7d19…`). Over the measurement window (13:14:41 to 14:14:41 London) it completed 67.1 calls a minute, and K_frozen = 108: 108 fits, 120 fails, and the analysed positions are 1–108. The computed cut time is 21 October 23:39 London, and the projected completion with margin is 23 October 13:31 London, against the 18:00 stop (`k-frozen.json`; `k-freeze-verification-v3.json`, which reads no outcome fields). All 108 analysed positions ran between 19 and 23 October; 106 worlds are complete and 2 partial (positions 4 and 40). The accepted coverage record counts 250,600 physical model starts. Positions 99–108 were dispatched after the computed cut time of 21 October 22:39 UTC; they are analysed, and a post-collection sensitivity masks them (Section 7.6).

W3.7 Analysis

The matched contrasts D_S, D_I and Θ are computed for each outcome and case over the scheduled worlds, with missing outcomes bounded within the fixed denominators [Ada SAP §§2–5]. We report means, ranges, distributions across worlds and bounds. This was the reporting position at launch: no p-values on simulated contrasts. After collection, the author activated the conditional procedure for the 90 contrasts, the two-factor analysis of exploration added 18 conditional sensitivity tests (Section S-F), and the capacity model added nine conditional tests (Section 6.6). The witness statement is reported on the K_frozen prefix (MF-S7).

W3.8 Reproducibility and nondeterminism

Hosted models are not deterministic at temperature zero. In attempt 3b, 16 Luna analysis requests sent twice with byte-identical bodies (once in calibration, once in development Q4) returned identical text in 4 and the same per-world verdict in 10. Toby established this by comparing saved responses. Luna carries no seed; the Qwen seed derives from the call identity. Within each design the round-5 state is copied into both price branches. Each world's round-0 state is shared by all four branches; rounds 1–5 run separately for each design. Frozen manifests are hashed (attempt 5: `74955b56…`; study v2: `ad1b2226…`; launch: `962d7d19…`, differing from v2 by three runner pins; successor runner: `02bdebdc…`). The trace is hash-chained, every request body is recorded, and every gate was recomputed from saved native responses. Qualification ran at 8 workers and the study at 48; no effect of load on answers is known, and none has been measured.

W3.9 Live-run interface rulings

All of these rulings were made under the repair-and-go rule, as interface repairs. None changed a bank, prompt, pay, seed, world order, K rule or gate.

- **Launch wiring (Addendum 3, before launch).** One world's failure or pause no longer halts the study: its unfinished positions become NA and every other lane continues. A funding pause or coordinator stop halts new dispatch and preserves in-flight work. Resume reuses the same run directory, chain, K_frozen and cursor, and never re-freezes K. The unstarted-at-cut flag is written at once.
- **Crash recovery and memory (Addendum 4).** A crash is a halt, not a cancellation. In-flight calls become NA with the cause "process crash", and recovery resumes from the durable chain and cursor. Resident memory is monitored against a 32 GiB guard.
- **Trace spill (Addendum 5).** The trace had been held in memory and was growing. An additive on-disk trace and call index preserves the exact chain bytes. The switch is one controlled pause, recorded at 15:16 London on 19 October (`pause-00001.json`, cause "interface repair: trace spill"), and a resume onto the successor runner from the same directory, chain, cursor and K_frozen. Resume time and pause duration are not reported in the sources used for this draft.
- **Provider budget overruns (Addendum 6).** Twice in the first hour the worker's provider returned more reasoning tokens than its declared 4,096 budget (4,097 once; 8,192 once). The old runner treated the resulting cap-hit answer as a world failure. Under the frozen scoring rules, a cap hit or invalid answer on a task is an observed wrong answer for that task, and invalid decisions, analyses and adviser outputs follow their frozen fallbacks. The world-level classification is a runner defect. Worlds ended by it resume from their cursor after the switch, with their settled calls standing. Worlds 9 and 26 are named in the ruling. Five halt reports (`LEO-REPORT-ANALYSED-UNSTARTED-AT-HALT-world_failure-048` to `-052`) were written between 13:41 and 14:48 London, and Avery lists the affected worlds at the switch. The accepted output records 53 native overruns and 3 historical recoveries as administered treatment (class 4 events), which are not masked; counts by role and arm are not reported here.

S-K. Exploratory analysis of exploration over time

This appendix supports Section 8.2. The analysis was specified after the outcomes and the analyses in Section 7 had been seen, and after an earlier diagnostic of the same saved data. It is descriptive: it reports counts and equal-world means, no tests and no intervals, and none of its quantities enters the estimates in Section 7. It uses the 106 complete worlds and 424 branches of the primary analysis (12,720 branch-rounds) with the same exclusions. Rounds 1–5 are one realization per world and technology, copied into the shock and no-shock branches; their rows are identical by construction and are never counted twice as evidence. Rounds, events and branches are repeated observations of the same worlds, not independent workers.

S-K.1 Definitions and cash accounting

Exploration (b) is the purchased exploration quantity in a round, on a quarter-unit grid: observations of one initially unfamiliar target domain plus an AI analysis of them. It omits unpaid workroom practice and what paid work on adjacent inputs reveals, so it does not measure every learning activity. *Participation* is whether $b > 0$ in a round. *Assistance (a)* is purchased production help on paid work, as in Section 6.4; it is not the later use of a newly acquired domain. *A* is defined as in Section 6.4, the share of scheduled initially unfamiliar domain-rounds served successfully, with a fixed denominator of three or four domains per round; it is neither the number of domains ever discovered nor ability without AI. Block means are formed within each world and then averaged with equal world weights over six fixed blocks, rounds 1–5, 6–10, 11–15, 16–20, 21–25 and 26–30.

Timing. The worker chooses paid work x, assistance a and exploration b together at the start of a round. Paid tasks are answered first, and the purchased observations and analysis arrive afterwards, so a purchase can inform answers only in later rounds. *Earlier* always means a strictly earlier round.

Cash. Purchases are paid from opening cash w, so $p(a + b) \leq w$. At settlement the worker holds $S = w - p(a + b) + Y + (1 - p)$, where Y is the round's earnings from accepted paid scores and $1 - p$ the rebate; she consumes $S/2$ and carries $S/2$ forward, and in round 30 she consumes S. Opening cash in round 1 is 2. Without the shock $p = 1$ and the rebate is zero. With it, $p = 1/4$ from round 6 and the rebate is $3/4$ per round, 18.75 credits over rounds 6–30, and the price of exploration relative to assistance is unchanged. Because $S \geq 3/4$ whenever spending stays within opening cash, carried cash on the shock path is at least $3/8$ from round 7. The value $3/4$ is the limit of carried cash with no spending and no earnings, not a floor for any finite round, and round 6 can inherit low cash from the shared history.

Affordability. The minimum purchase $b = 1/4$ costs $p/4$: 0.25 without the shock and 0.0625 with it. It is affordable *before assistance* if $w \geq p/4$ and *after assistance* if $w - pa \geq p/4$, where a is the assistance chosen in the same round. Because a and b are chosen together, the second measure subtracts a jointly chosen quantity; it is not an observed sequence of purchases. Neither measure uses the round's earnings, its rebate or later cash. We say that a worker *could not afford* exploration only when $w < p/4$.

Time for exploration. Paid work $x > 0$ carries a fixed overhead of a quarter round, and a positive exploration purchase needs a further quarter, so $b > 0$ fits only when $x \leq 1/2$. With $x = 3/4$, the largest allowed, no exploration purchase fits. A round with no time for b describes the allocation the worker chose, not a constraint imposed on her. When a decision was invalid, the validator substituted $x = 3/4$, $a = 0$ and $b = 0$; we call these rows validator fallbacks, keep them, and treat them as neither a choice not to buy nor an inability to buy.

Successes and their links. An event is an initially unfamiliar domain-round in rounds 6–30 that counts toward A. It is linked to the first and the most recent strictly earlier purchase in the same domain, and the lag is the difference in rounds. L is the branch's last round with $b > 0$, and the domain's final purchase is the last round in which that domain was the target; the two are reported separately. Remaining follow-up after L is $30 - L$. Links between purchases and successes overlap, because one success can follow several purchases and one purchase can precede several successes.

Answer origin and card state. Origin follows Section S-C: the worker's own correct answer first, then an adopted helper answer in a domain whose complete card she held (known-domain helper), then an adopted helper answer otherwise (latent-domain helper). Card state is read before the answer: no card; present

but structurally incomplete; structurally complete; and full-domain correct, which the investigator checks against the hidden mechanism over all 8,192 inputs of the domain. Card states describe availability. They do not show that a card produced the answer, and the record does not show where a card's content came from.

Stopping. Branches are never-purchasers, one-time purchasers and repeat purchasers. A purchaser stops if $L < 30$, and $\tau = L + 1$ is the round at which cash, time and card states are read. Prior outcomes are the investigator-observed paid attempts in the worker's purchased target domains between her first purchase and L. The worker herself saw her aggregate earnings and, every five rounds, an aggregate certification share, never her results by domain. A full record does not prevent another purchase, because every initially unfamiliar domain remains an eligible target.

S-K.2 Joint trajectories

Table S-K1 gives the pooled trajectories, and Tables S-K2 and S-K3 give exploration and task expansion by technology and propensity so that pooling cannot conceal a difference between the two designs. Round-level trajectories, unsmoothed, are in the supporting data file named in S-K.7. In the shock branch, purchases did not decline steadily: the pooled series rose in rounds 21–25 before falling in rounds 26–30, and several cells in Table S-K2 rose late.

Table S-K1. Joint trajectories by five-round block, both technologies and all propensity groups. Equal-world means over 106 complete worlds; each block and path has 1,060 branch-rounds and 3,720 scheduled initially unfamiliar domain-rounds. Rounds 1–5 are the shared pre-shock history and are identical in both rows by construction. Affordability and time are defined in S-K.1; validator fallbacks remain in every row.

Measure	Path	1–5	6–10	11–15	16–20	21–25	26–30
Exploration b, units per round	Shock	0.0104	0.0101	0.0073	0.0064	0.0080	0.0047
Exploration b, units per round	No shock	0.0104	0.0014	0.0012	0.0009	0.0014	0.0014
Participation, % of branch-rounds	Shock	4.15	3.68	2.92	2.45	2.83	1.60
Participation, % of branch-rounds	No shock	4.15	0.47	0.47	0.38	0.57	0.57
Assistance a, units per round	Shock	0.057	0.054	0.060	0.050	0.054	0.046
Assistance a, units per round	No shock	0.057	0.008	0.009	0.008	0.008	0.007
A, % of scheduled unfamiliar domain-rounds	Shock	0.22	0.92	1.46	1.62	2.01	2.34
A, % of scheduled unfamiliar domain-rounds	No shock	0.22	0.71	0.58	0.42	0.61	0.56
Minimum b affordable before a, %	Shock	73.9	88.8	100.0	100.0	100.0	100.0

Measure	Path	1–5	6–10	11–15	16–20	21–25	26–30
Minimum b affordable before a, %	No shock	73.9	7.7	6.9	7.5	8.2	6.8
Minimum b affordable after the chosen a, %	Shock	65.2	86.4	100.0	100.0	100.0	100.0
Minimum b affordable after the chosen a, %	No shock	65.2	5.0	4.0	5.1	5.6	4.8
Chosen allocation leaves time for b, %	Shock	29.2	28.6	19.3	20.4	21.9	15.7
Chosen allocation leaves time for b, %	No shock	29.2	28.7	27.8	24.4	24.8	27.3
Validator fallbacks, branch-rounds	Shock	6	9	6	7	4	7
Validator fallbacks, branch-rounds	No shock	6	2	3	0	3	5

Table S-K2. Exploration b, units per round, by technology, propensity and five-round block. Equal-world means; 34 low-, 36 medium- and 36 high-propensity worlds. Rounds 1–5 are shared within a technology.

Technology	Propensity	Path	1–5	6–10	11–15	16–20	21–25	26–30
Both	All	Shock	0.0104	0.0101	0.0073	0.0064	0.0080	0.0047
Both	All	No shock	0.0104	0.0014	0.0012	0.0009	0.0014	0.0014
Both	Low (λ = 0)	Shock	0.0088	0.0044	0.0044	0.0015	0.0044	0.0007
Both	Low (λ = 0)	No shock	0.0088	0.0007	0.0015	0.0007	0.0015	0.0000
Both	Medium (λ = 1/2)	Shock	0.0111	0.0118	0.0090	0.0056	0.0063	0.0076
Both	Medium (λ = 1/2)	No shock	0.0111	0.0021	0.0000	0.0007	0.0000	0.0021
Both	High (λ = 19/20)	Shock	0.0111	0.0139	0.0083	0.0118	0.0132	0.0056
Both	High (λ = 19/20)	No shock	0.0111	0.0014	0.0021	0.0014	0.0028	0.0021
Personalized	All	Shock	0.0123	0.0123	0.0080	0.0075	0.0080	0.0057
Personalized	All	No shock	0.0123	0.0009	0.0014	0.0019	0.0019	0.0028
Personalized	Low (λ = 0)	Shock	0.0074	0.0059	0.0029	0.0029	0.0015	0.0015
Personalized	Low (λ = 0)	No shock	0.0074	0.0000	0.0015	0.0015	0.0015	0.0000
Personalized	Medium (λ = 1/2)	Shock	0.0167	0.0181	0.0125	0.0042	0.0069	0.0056
Personalized	Medium (λ = 1/2)	No shock	0.0167	0.0028	0.0000	0.0014	0.0000	0.0042
Personalized	High (λ = 19/20)	Shock	0.0125	0.0125	0.0083	0.0153	0.0153	0.0097
Personalized	High (λ = 19/20)	No shock	0.0125	0.0000	0.0028	0.0028	0.0042	0.0042
Integrated	All	Shock	0.0085	0.0080	0.0066	0.0052	0.0080	0.0038

Technology	Propensity	Path	1–5	6–10	11–15	16–20	21–25	26–30
Integrated	All	No shock	0.0085	0.0019	0.0009	0.0000	0.0009	0.0000
Integrated	Low (λ = 0)	Shock	0.0103	0.0029	0.0059	0.0000	0.0074	0.0000
Integrated	Low (λ = 0)	No shock	0.0103	0.0015	0.0015	0.0000	0.0015	0.0000
Integrated	Medium (λ = 1/2)	Shock	0.0056	0.0056	0.0056	0.0069	0.0056	0.0097
Integrated	Medium (λ = 1/2)	No shock	0.0056	0.0014	0.0000	0.0000	0.0000	0.0000
Integrated	High (λ = 19/20)	Shock	0.0097	0.0153	0.0083	0.0083	0.0111	0.0014
Integrated	High (λ = 19/20)	No shock	0.0097	0.0028	0.0014	0.0000	0.0014	0.0000

Table S-K3. Task expansion A, percent of scheduled initially unfamiliar domain-rounds served successfully, by technology, propensity and five-round block. Equal-world means with fixed initial-domain denominators.

Technology	Propensity	Path	1–5	6–10	11–15	16–20	21–25	26–30
Both	All	Shock	0.22	0.92	1.46	1.62	2.01	2.34
Both	All	No shock	0.22	0.71	0.58	0.42	0.61	0.56
Both	Low (λ = 0)	Shock	0.27	0.98	1.00	1.32	1.62	1.15
Both	Low (λ = 0)	No shock	0.27	0.64	0.22	0.34	0.56	0.37
Both	Medium (λ = 1/2)	Shock	0.16	0.79	1.83	2.15	1.44	2.31
Both	Medium (λ = 1/2)	No shock	0.16	0.95	0.74	0.83	0.72	0.65
Both	High (λ = 19/20)	Shock	0.23	1.00	1.53	1.37	2.96	3.50
Both	High (λ = 19/20)	No shock	0.23	0.53	0.76	0.07	0.53	0.65
Personalized	All	Shock	0.38	1.10	2.23	2.17	2.22	2.30
Personalized	All	No shock	0.38	0.75	0.75	0.50	0.69	0.58
Personalized	Low (λ = 0)	Shock	0.34	0.93	1.27	1.08	1.42	0.83
Personalized	Low (λ = 0)	No shock	0.34	0.29	0.15	0.15	0.69	0.15
Personalized	Medium (λ = 1/2)	Shock	0.32	1.25	2.92	3.24	1.34	1.76
Personalized	Medium (λ = 1/2)	No shock	0.32	1.06	1.16	1.20	0.32	0.60
Personalized	High (λ = 19/20)	Shock	0.46	1.11	2.45	2.13	3.84	4.21
Personalized	High (λ = 19/20)	No shock	0.46	0.88	0.93	0.14	1.06	0.97
Integrated	All	Shock	0.06	0.74	0.69	1.07	1.81	2.39
Integrated	All	No shock	0.06	0.66	0.41	0.33	0.52	0.53
Integrated	Low (λ = 0)	Shock	0.20	1.03	0.74	1.57	1.81	1.47
Integrated	Low (λ = 0)	No shock	0.20	0.98	0.29	0.54	0.44	0.59
Integrated	Medium (λ = 1/2)	Shock	0.00	0.32	0.74	1.06	1.53	2.87
Integrated	Medium (λ = 1/2)	No shock	0.00	0.83	0.32	0.46	1.11	0.69
Integrated	High (λ = 19/20)	Shock	0.00	0.88	0.60	0.60	2.08	2.78
Integrated	High (λ = 19/20)	No shock	0.00	0.19	0.60	0.00	0.00	0.32

S-K.3 Cash and time after the shock

Table S-K4 separates three conditions that Section 8.2 keeps apart: whether opening cash covered the minimum purchase, whether it still did after the assistance chosen in the same round, and whether the chosen

allocation left time for exploration. Affordability in rounds 6–10 of the shock branch is 88.8 rather than 100 percent because round 6 inherits cash from the shared history. Without the shock, opening cash fell after the early rounds and stayed low; the lowest opening cash in rounds 7–30 is zero.

Table S-K4. Cash and time for the minimum exploration purchase after the shock (branch-round counts; 106 worlds, 212 branches per path). The minimum purchase b = 1/4 costs 0.25 without the shock and 0.0625 with it. Opening cash excludes the round's own earnings and rebate. A round with no time for b reflects the allocation the worker chose together with b.

Rounds 6–30	Shock	No shock
Branch-rounds, rounds 6–30	5,300	5,300
Minimum b affordable before a	5,181 (97.8%)	393 (7.4%)
Minimum b affordable after the chosen a	5,156	259
Rounds with b > 0	143	26
Affordable before a, no purchase	5,038	367
Of which: affordable after a, valid decision, no time for b (x = 3/4)	4,041	181
Of which: affordable after a, valid decision, time for b available	939	46
Of which: not affordable after a	25	134
Of which: validator fallbacks	33	6
Minimum b not affordable before a	119	4,907
Validator fallbacks, all rounds 6–30	33	13
Lowest opening cash, rounds 7–30	3/8	0

S-K.4 Successful unfamiliar work and earlier purchases

The links in Tables S-K5 and S-K6 are temporal associations. They show that a purchase in the same domain came first, not that it supplied the answer. Without the shock, most links run to purchases made in rounds 1–5: 74 of the 101 linked events had no later same-domain purchase, and 77 of the 91 successes after the branch's final purchase came with 20 or more rounds of follow-up remaining.

Table S-K5. Successful initially unfamiliar domain-rounds and their links to earlier exploration purchases. Repeated-event counts, not independent observations: the ten most active branches contribute 106 of the 417 events and one world 33. Lags are in rounds; quartiles use the nearest rank. Routes use the own-first priority of Section S-C; 15 events with both an own and a known-domain helper correct answer are shown as own.

Events	All	Shock	No shock
Successful initially unfamiliar domain-rounds, rounds 6–30	417	310	107
Worlds	81	78	34
Branches	130	93	37
Preceded by a same-domain purchase in a strictly earlier round	388	287	101
Earlier purchases in other domains only	5	5	0
No earlier purchase	24	18	6
First same-domain purchase in rounds 1–5	197	116	81
Most recent same-domain purchase after round 5	247	220	27
Lag from first purchase, median (quartiles)	12 (7–19)	13 (7–19)	12 (7–20)
Lag from most recent purchase, median (quartiles)	9 (4–15)	8 (4–13)	11 (6–18)
After the branch's final purchase	322	231	91

Events	All	Shock	No shock
After the domain's final purchase	321	229	92
Route: own answer	373	272	101
Route: known-domain helper	17	16	1
Route: latent-domain helper	27	22	5

Table S-K6. Distribution of the lag between a success and the first or the most recent strictly earlier purchase in the same domain (388 linked events; 287 with the shock, 101 without). The maximum lag is 28 rounds.

Lag in rounds	First: all	First: shock	First: no shock	Latest: all	Latest: shock	Latest: no shock
1–5	68	51	17	126	104	22
6–10	93	66	27	108	84	24
11–15	80	63	17	65	48	17
16–20	70	52	18	43	25	18
21–25	55	42	13	32	21	11
26–28	22	13	9	14	5	9

S-K.5 Stopping

Table S-K7 keeps never-, one-time and repeat purchasers apart and reports the remaining follow-up for each stop, because a stop near round 30 leaves little time in which another purchase could pay and is not comparable with an early stop. The same stop can fit more than one account, and we assign none. A cash shortfall at τ is common without the shock (44 of 58 stops) and rare with it (15 of 113, all one-time purchasers). A bounded reading of the saved stated reasons for 23 decision sequences, sampled mechanically across propensity, technology, price path and early or late final purchase, found explicit statements of inability at zero cash, mostly in the shared rounds, and statements citing pay or advice when purchases stopped while affordable; none cited disappointing returns, a wrong card or having explored every domain, and none referred to the remaining horizon explicitly. The sample is not representative, and the texts are generated summaries, not reports of internal motives.

Table S-K7. Branches that purchased exploration, by price path and purchaser class. Never-purchasers, 99 with the shock and 153 without, are excluded. L is the last round with $b > 0$ and $\tau = L + 1$ the first round without one. Prior outcomes are investigator-observed paid attempts in purchased target domains between the first purchase and L; one-time purchasers have no such rounds by construction. Each card count has its own definition. After L, the last purchased target domain held a full-domain correct card in 6 of the 171 stops, a structurally complete but not correct card in 133 and no card in 32.

Purchasers	Shock, one-time	Shock, repeat	No shock, one-time	No shock, repeat	Total
Branches	66	47	48	11	172
Stopped before round 30	66	47	47	11	171
Purchased in round 30 (censored)	0	0	1	0	1
Last purchase in rounds 1–5 (shared history)	24	1	34	1	60
Remaining follow-up 30 − L: 1–4 rounds	4	12	2	3	21
Remaining follow-up: 5–9 rounds	6	10	3	2	21

Purchasers	Shock, one-time	Shock, repeat	No shock, one-time	No shock, repeat	Total
Remaining follow-up: 10–19 rounds	17	20	4	4	45
Remaining follow-up: 20–29 rounds	39	5	38	2	84
At τ: minimum b not affordable before a	15	0	34	10	59
At τ: affordable before a only	3	0	5	0	8
At τ: affordable before and after a	48	47	8	1	104
At τ: chosen allocation leaves time for b	16	12	15	2	45
Before stopping: no attempt in a purchased domain	66	11	47	1	125
Before stopping: attempts, no success	0	18	0	6	24
Before stopping: at least one success	0	18	0	4	22
After L: all five card slots filled	2	2	0	1	5
After L: structurally complete card for every initially unfamiliar domain	0	1	0	0	1
After L: full-domain correct card for every initially unfamiliar domain	0	0	0	0	0

S-K.6 Card correctness and follow-through

Card correctness has three distinct meanings here. A structurally complete card passes the record's structural check. A full-domain correct card reproduces the hidden mechanism on all 8,192 inputs. A correct answer on one task requires neither, and 320 of the 373 own successes came with a card that was complete but not correct across the domain. We did not check whether each card predicts the answer to the particular task it accompanied, so the tables report card availability, not answers produced by a card. All 417 card classifications were reproduced independently from the pinned checker definitions, for 88 functionally distinct cards, with no mismatch. No accepted task score was changed.

Table S-K8. Card state before the answer, by displayed route, for the 417 successful initially unfamiliar domain-rounds. Full-domain correctness is the investigator's check of the card against the hidden mechanism over all 8,192 inputs of the domain; the worker never saw it.

Card before the answer	Own answer	Known-domain helper	Latent-domain helper	All
Full-domain correct	51	0	0	51
Structurally complete, not full-domain correct	320	17	0	337
Present, structurally incomplete	0	0	1	1

Card before the answer	Own answer	Known-domain helper	Latent-domain helper	All
No card	2	0	26	28

Table S-K9. Follow-through after each purchase of exploration. A purchase-success link counts every later success in the purchased domain, so links overlap across purchases; rounds after 30 are unobserved, not failures. The 44 purchases made in rounds 1–5 are one realization each, listed under both paths.

Purchases	All	Shock	No shock
Branch-round purchases of exploration	257	187	70
Of which: made in rounds 1–5 (shared, listed under both paths)	88	44	44
No success in the target domain in any later round	95	61	34
Five later rounds observed	234	170	64
Of which: no success in the target domain in those five rounds	148	103	45

S-K.7 Reproduction

The inputs are the saved results of the 106 complete worlds of the live study (`live-study-v3/world-NNN/result.json`, with worlds 4 and 40 excluded as in the primary analysis), the frozen engine and context files (`hd5_study_engine_v7.py`, SHA-256 145c5c04…; `hd5_study_context.py`, fc88c0a1…), the pinned bank and checker definitions, and the saved diagnostic records of 25 October (`exploration-diagnostic-20261025/`). Avery reconstructed every round from the raw ledgers in memory, without executing the diagnostic scripts or importing project modules, and reproduced all eleven preliminary benchmarks, including the 2,544 accepted operand values behind A, a and b in Section 7. Scully recomputed the cohort, events, cash states, stopping counts, card classifications and every block cell in Tables S-K1–S-K3, which are also the values plotted in Figure S-M5, with separately written code; all 277,359 table fields matched. The field `following_target_successes` in the earlier diagnostic's `purchase-followthrough.csv` counts successful attempts rather than successful domain-rounds; Table S-K9 keeps the two apart. No study call, task, score or source file was added or changed. The figure data, round-level trajectories and plotting code are in `icis-exploration-figures-2026-10-25/` beside this manuscript.

S-M. Main-text material moved for the page limit

Moved on 28 October 2026 from the 25 October draft to keep the main paper within eight pages. Content is unchanged; tables and figures keep their 25 October numbers with the prefix S-M, and pointers to the former appendices now name the supplement sections. Section numbers in this part refer to the paper unless marked.

S-M.1 Design and operationalisation (Sections 6.1–6.2 of the 25 October draft)

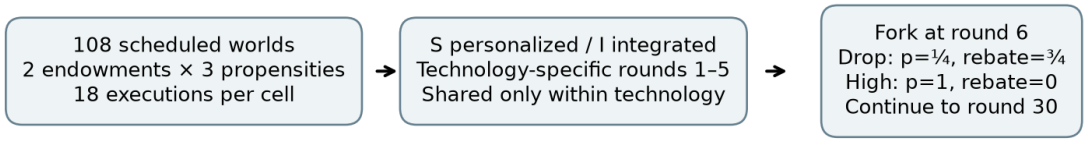


Figure S-M1. The design. Each world is run under two technologies, which diverge from round 1; within each technology, the shock and no-shock branches share rounds 1–5 and fork at round 6. Primary analysis: 106 complete worlds.

Each domain maps an integer u from 0 to 1023, an integer v from 0 to 3 and a category $c \in \{A, B\}$ to an integer. Only the checker holds the mechanism. Paid tasks ask for the output at given inputs or, in hard domains, for the u that produces a stated target. A wrong answer earns nothing. Two domains are easy, with an affine mechanism of four constants. Three are hard: two follow a step form and one a monotone two-step form, each with eight constants and a threshold limited to two separated values. A seeded generator draws the constants, so no prior can guess them. Every model payload carries the same public grammar of forms and ranges, with no constant of any world. Each quarter of b buys two observations, and the first five in each domain identify an easy mechanism exactly. Pay is fixed per domain and was scaled once before launch, at $s = 1/2$.

The worker is Qwen3.7-flash, with reasoning on, a 4,096-token thinking budget and temperature zero. The advisers, the helper and the analysis service are GPT-6 Luna at low effort. Her record holds at most five cards; numbers may appear only in typed parameter fields, and task identifiers and answers are banned. Because the model's weights never change, capability means fresh performance supported by a text record (Shinn et al. 2023). Section S-G gives the full model and Section S-H every design parameter.

Before any treatment data existed, six gates certified the instrument (Table S-M1). All were frozen before any paid call, and none reads a technology or price contrast. Seven paid runs preceded the pass, under one rule set by the author: an interface failure is repaired prospectively and rerun under a new number, while a frozen gate that is reached and fails is a result. Three design amendments followed reached failures, each before any treatment branch had run; Section S-H reports every run, cost and amendment, including the case for admitting the third.

Table S-M1. Qualification gates and the result of attempt 5 (19 October 2026)

Gate	What it certifies	Attempt 5
Q1 Oracle	A true card can be applied	Easy 63/64; hard 62/64

Gate	What it certifies	Attempt 5
Q2 Empty ceiling	Priors alone cannot answer	0/64 in each band
Q3 Thinking room	The budget is not binding	0/64 calls at the budget
Q4 Inferability	Exploration can succeed and fail	Fresh: easy 23/24; hard 11/16 succeed, 5 fail
Q5 Helper	Production help is accurate	Easy 59/64; hard 60/64
Q6 Acquisition	An inferred record lets the worker answer fresh tasks	Fresh: 91/96 with record vs 0/96 without

S-M.2 Estimands and model (Section 6.5 of the 25 October draft)

For world w and technology d, let Y_{pre} be the per-round outcome in rounds 1–5 and Y_{post} the same outcome in rounds 6–30, in the shock branch (drop) and the no-shock branch (high). We define five world-level contrasts:

$$DiD_{w,d} = \left(Y_{post}^{drop} - Y_{pre}^{drop}\right) - \left(Y_{post}^{high} - Y_{pre}^{high}\right),$$

$$D_w = \tfrac{1}{2}\left(DiD_{w,S} + DiD_{w,I}\right), \qquad X_w = DiD_{w,S} - DiD_{w,I},$$

$$T_w = \tfrac{1}{2}\left(Y_{post}^{S,drop} + Y_{post}^{S,high} - Y_{post}^{I,drop} - Y_{post}^{I,high}\right), \qquad M_w = \tfrac{1}{4}\sum_{d,p} Y_{post}^{d,p}.$$

Because the two price branches share rounds 1–5 within a technology, $Y_{pre}^{drop} = Y_{pre}^{high}$ and the difference-in-differences reduces to the matched post-period contrast, $DiD_{w,d} = Y_{post}^{drop} - Y_{post}^{high}$. D is the shock's effect averaged over the two technologies, X is how much larger it is under the personalized design, T is the personalized-minus-integrated difference in the post-period level averaged over both price paths, and M is the post-period level averaged over all four branches.

For each headline outcome, B and A, we report seven effects in one factorial presentation. The overall shock effect is the mean of D; the technology level difference is the mean of T; the shock-by-technology interaction is the mean of X. The propensity level difference, the technology-by-propensity interaction, the shock-by-propensity interaction and the three-way interaction are the joint comparisons of M, T, D and X across the three propensity groups. Each joint comparison has two components, medium minus low and high minus low, which we also report. The same quantities are the cell contrasts of a post-period model, $Y_{post} \sim$ Drop × Technology × Propensity, with all lower-order terms and propensity as a category. In the before/after panel, $Y_{phase} \sim$ Post × Drop × Technology × Propensity, the shock is the Post × Drop term, so the shock-by-technology-by-propensity interaction is a four-way term in that notation (Section S-A gives the mapping, the full coefficients and the covariance matrices). The reference-coded coefficients of either model are effects at the reference categories, not average effects, and we do not interpret them as such.

Inference. Each scalar effect is tested with a two-sided one-sample t test on the 106 world-level contrasts (df = 105), with a pointwise 95% interval. Each joint comparison is a one-way analysis of variance across the three propensity groups (df = 2, 103), whose medium-minus-low and high-minus-low coefficients carry intervals based on the pooled residual variance. Within a propensity group, we use paired world-level t intervals with that group's degrees of freedom (33, 35 and 35). The paired t test and the model-based test of the same contrast give slightly different p-values, because the first uses the variance of the contrast across all worlds and the second the pooled within-group variance; Section S-A reports both, and none of the conclusions changes.

Multiplicity. The headline family has fourteen tests, the seven effects for each of B and A, and we report Holm-adjusted p-values beside the raw ones. A second, separate family of seven supporting tests covers exploration participation (the overall technology difference, the propensity level difference, the technology-by-propensity interaction and the three technology contrasts within propensity groups) and the overall shock effect on production assistance. We did not select outcomes by significance, and the earlier families of 90, 18 and nine tests keep their own adjustments in Sections S-D to S-F.

Two cautions about reading the effects. First, because the branches share their pre-shock history, the post-period level of a world is its no-shock level plus half the shock effect, $M = M_{high} + D/2$, and likewise $T = T_{high} + X/2$. A propensity level difference and a shock-by-propensity interaction can therefore partly report the same thing, and we do not count them as independent evidence. Second, the tests treat worlds as independent stochastic executions. The worlds share a task bank, prompts, models, two endowment records and provider epochs, and the replicates within a case share their record and λ. Rounds, cloned prefixes and branches are never treated as independent workers. Whether the nominal t and F reference distributions are calibrated for this bank is unknown. The intervals and p-values describe repeated execution of this synthetic design, not a human population.

Robustness. Section S-B reports, without additional tests, effects standardized to equal endowment weights within propensity, world-resampled bootstrap intervals that keep all four branches of a world together, record-adjusted propensity tests, and the scheduled-108 missingness bounds.

Exploratory addition. After the results in Section 7 had been seen, we added a descriptive analysis of how exploration purchases, cash and successfully served unfamiliar work moved together across the 30 rounds, in the same 106 worlds and with the outcomes defined above. It adds no tests and changes no estimate; Section 8.2 of the paper, Figure S-M5 and Section S-K report it.

S-M.3 Tables and figures moved from Section 7

Table S-M2. Before and after the shock: period means over 106 equally weighted complete worlds, both technologies pooled. B is purchased exploration per round; A is the percent of scheduled initially unfamiliar domain-rounds served successfully. Rounds 1–5 are the same history in both price branches. The difference-in-differences (DiD) equals the post-period shock-minus-no-shock contrast.

Outcome	Rounds 1–5	Rounds 6–30, no shock	Rounds 6–30, shock	Change, no shock	Change, shock	DiD [95% CI]
B, exploration per round	0.0104	0.0013	0.0073	−0.0091	−0.0031	+0.0060 [0.0046, 0.0075]
A, task expansion (%)	0.22	0.57	1.67	+0.35	+1.45	+1.10 [0.70, 1.50]

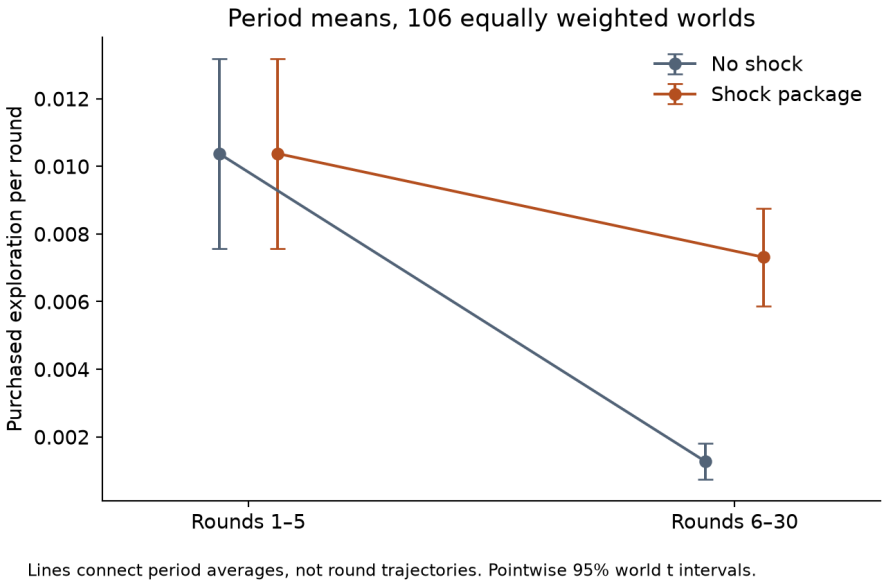


Figure S-M2. Exploration quantity B, purchased exploration per round, before and after the shock. Points are period means over 106 equally weighted complete worlds with pointwise 95% world t intervals. Lines connect the two period averages; they are not round-by-round trajectories.

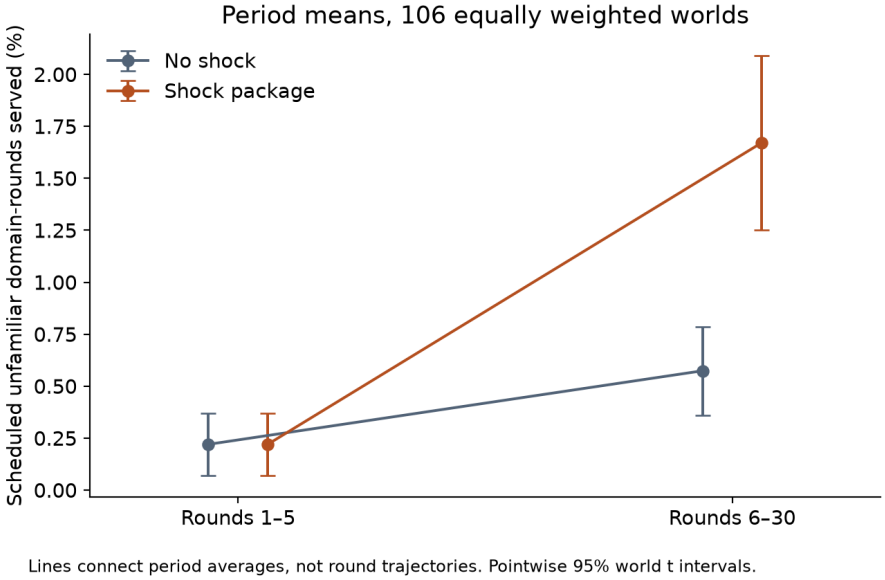


Figure S-M3. Task expansion A, percent of scheduled initially unfamiliar domain-rounds served successfully with any AI help the worker chose, before and after the shock. Symbols as in Figure S-M2; lines connect period averages, not round trajectories.

Table S-M4. Where the shock acted, and technology differences by propensity (106 complete worlds). Shock effects are mean difference-in-differences; technology differences are post-period S − I levels averaged over both price branches. Paired world t intervals, df = 33 (low), 35 (medium, high) or 105 (all). These are views of the factorial model in Table 1 of the paper, not additional tests.

View	Group	B: estimate [95% CI]	A: estimate [95% CI]
Shock effect	Low propensity (n = 34)	+0.0022 [0.0005, 0.0039]	+0.79 [0.10, 1.48]
Shock effect	Medium propensity (n = 36)	+0.0071 [0.0045, 0.0096]	+0.93 [0.08, 1.77]
Shock effect	High propensity (n = 36)	+0.0086 [0.0058, 0.0115]	+1.56 [1.02, 2.10]
Shock effect	Personalized (S), all worlds	+0.0065 [0.0042, 0.0088]	+1.35 [0.71, 1.98]
Shock effect	Integrated (I), all worlds	+0.0056 [0.0035, 0.0076]	+0.85 [0.43, 1.27]
Technology, S − I	Low propensity	−0.0001 [−0.0016, 0.0013]	−0.25 [−1.04, 0.54]
Technology, S − I	Medium propensity	+0.0021 [−0.0013, 0.0054]	+0.49 [−0.44, 1.42]
Technology, S − I	High propensity	+0.0025 [−0.0016, 0.0066]	+0.97 [0.06, 1.87]

Table S-M5. Exploration participation J_b: percent of branches with any exploration purchase in rounds 6–30, averaged equally over the shock and no-shock branches (106 complete worlds). The relative increase is the S − I difference divided by the integrated rate. Paired world t intervals; Holm adjustment within the seven-test supporting family (Table S-M6).

Propensity	Personalized (S)	Integrated (I)	S − I, points [95% CI]	Relative to I	p	Holm p
Low (n = 34)	17.6%	17.6%	0.0 [−12.1, 12.1]	0.0%	1	1
Medium (n = 36)	34.7%	19.4%	+15.3 [−0.3, 30.8]	78.6%	.054	.217
High (n = 36)	40.3%	27.8%	+12.5 [−2.8, 27.8]	45.0%	.107	.321
All (n = 106)	31.1%	21.7%	+9.4 [1.2, 17.6]	43.5%	.025	.123

The joint test of whether the S − I participation difference varies with propensity gives F(2, 103) = 1.27, p = .285, Holm p = .569.

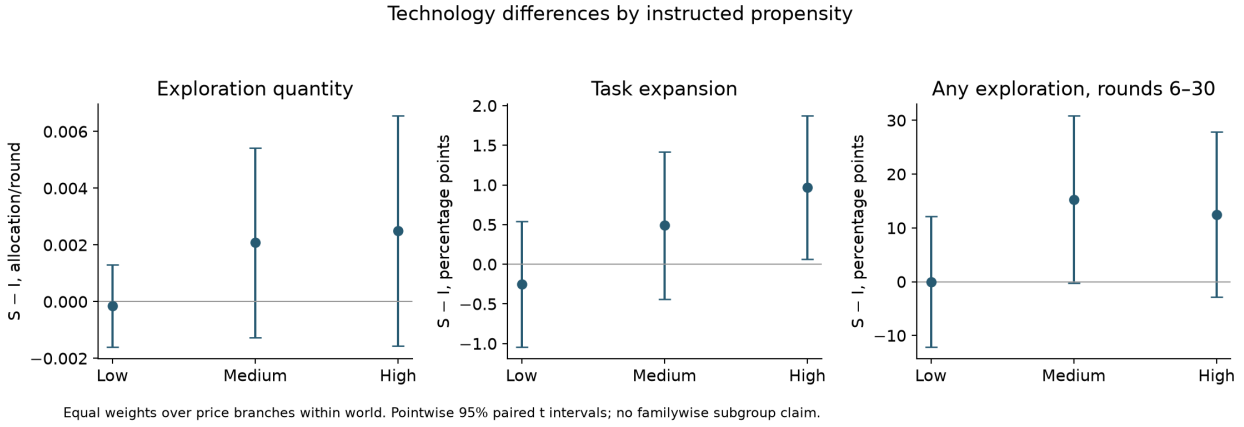


Figure S-M4. Technology differences (personalized minus integrated) by propensity, averaged over both price paths: exploration quantity per round, task expansion in percentage points, and exploration participation in rounds 6–30 in percentage points. Pointwise 95% paired t intervals; no familywise subgroup claim.

Table S-M6. The supporting family of seven tests, Holm-adjusted within the family. J_b in percentage points; a in purchased assistance per round, which is not completed output.

Outcome	Effect	Estimate [95% CI] or F	df	p	Holm p
Participation J_b	Technology, S − I, all worlds	+9.4 [1.2, 17.6]	105	.025	.123
Participation J_b	Propensity level difference	F = 5.24	2, 103	.0068	.041
Participation J_b	Technology × propensity	F = 1.27	2, 103	.285	.569
Participation J_b	S − I, low propensity	0.0 [−12.1, 12.1]	33	1	1
Participation J_b	S − I, medium propensity	+15.3 [−0.3, 30.8]	35	.054	.217
Participation J_b	S − I, high propensity	+12.5 [−2.8, 27.8]	35	.107	.321
Production assistance a	Overall shock	+0.0450 [0.0390, 0.0509]	105	5.2×10^{-28}	3.6×10^{-27}

Coefficients of the joint tests, in percentage points: propensity level, medium − low +9.4 [−0.6, 19.5] and high − low +16.4 [6.3, 26.4]; technology × propensity, medium − low +15.3 [−4.9, 35.4] and high − low +12.5 [−7.7, 32.7].

S-M.4 Exploration over time: Section 8.2 of the 25 October draft in full

In the shock branch, exploration and task expansion moved in opposite directions over time. Workers bought the most exploration in the first ten rounds, 0.0104 and 0.0101 units per round in rounds 1–5 and 6–10, and the least in rounds 26–30, 0.0047, with a partial recovery in rounds 21–25. Successfully served unfamiliar work, by contrast, rose in every five-round block, from 0.22 percent of scheduled unfamiliar domain-rounds in rounds 1–5 to 2.34 percent in rounds 26–30 (Figure S-M5). The two movements are compatible. A purchase of exploration buys observations of an unfamiliar mechanism and an analysis of them, and the card the worker writes from them can stay in her record and be used in any later round. A worker who explored a domain early can therefore keep serving it after she has stopped buying exploration, so fewer new purchases can coexist with more successful unfamiliar work. We examined whether the data fit this account, and the alternatives to it, after the results in Section 7 had been seen. The analysis is descriptive and exploratory, and it reports no tests (Section S-K).

The timing of purchases and successes is consistent with this account, but it does not establish it. In rounds 6–30, workers successfully served 417 initially unfamiliar domain-rounds, in 130 of the 424 branches and 81 of the 106 worlds. In 388 of them, the worker had bought exploration of the same domain in a strictly earlier round; paid work comes before exploration within a round, so a purchase in the same round could not have supplied that round's answer. The median success came 12 rounds after the first purchase in its domain and nine rounds after the most recent one. Of the 373 successes that were the worker's own answers, 371 followed such a purchase, and 322 of all 417 came after the branch's final purchase of exploration. Of the 29 successes with no earlier purchase in the same domain, 27 were helper answers that the worker adopted in a domain for which she held no complete card. Early acquisition alone does not account for the effect of the shock, however. The purchases made in rounds 1–5 are the same in both branches, yet a first purchase in those rounds preceded 116 later successes with the shock and 81 without it, and 220 of the 287 purchase-linked successes in the shock branch had their most recent same-domain purchase after round 5. A card was present before 371 of the 373 own successes, but only 51 of those cards were correct across the whole domain; the other 320 were complete in form and wrong for some inputs. A card can therefore support a correct answer on one task without being a correct account of its domain, and its presence does not show that it produced the answer. These successes demonstrate task capability acquired with the record and AI available. They do not show what the worker could do without live AI, which Section S-C measures separately and finds to be small.

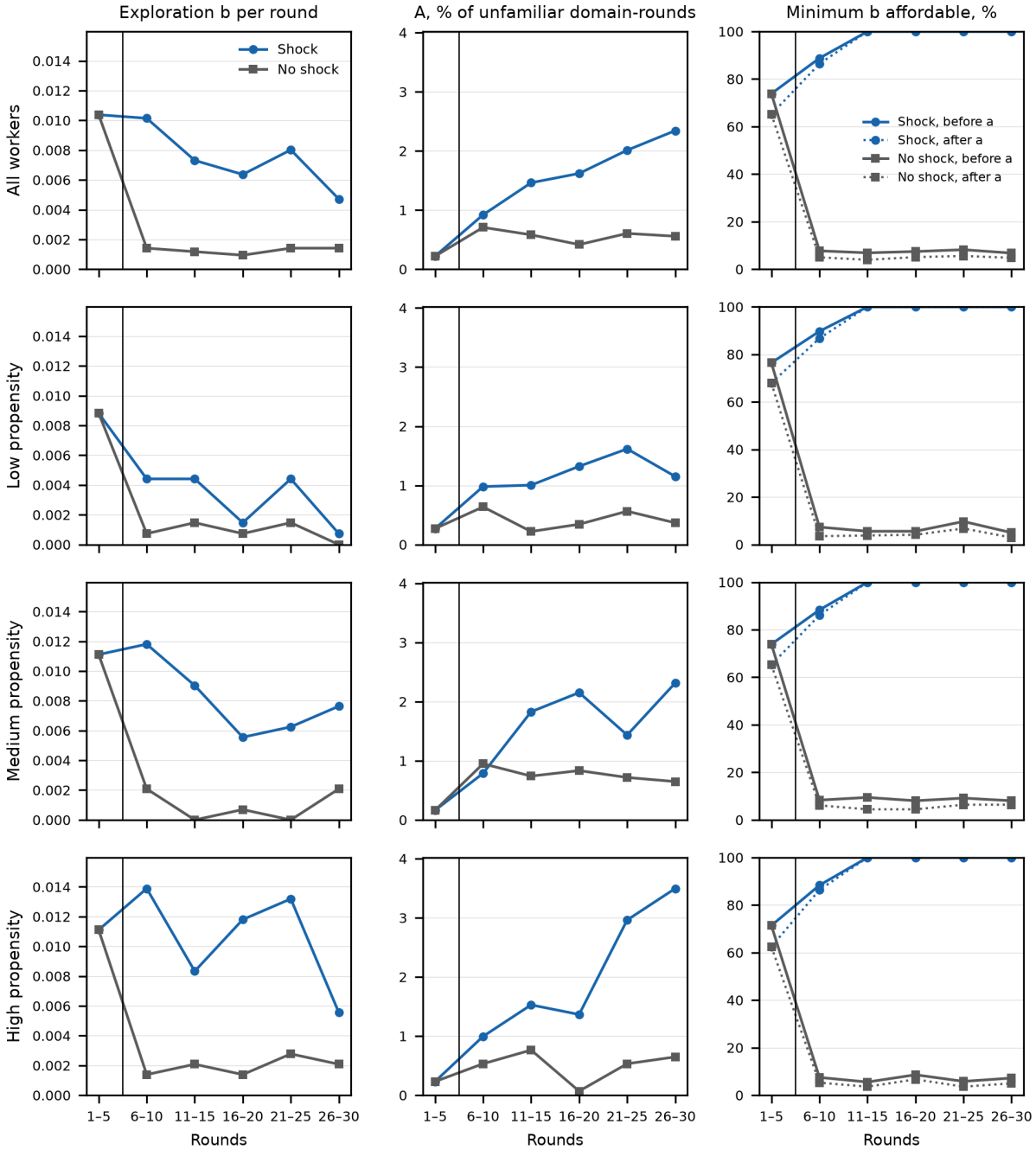


Figure S-M5. Exploration purchases, successfully served unfamiliar work and cash affordability by five-round block, price path and propensity. Columns: exploration b purchased per round; task expansion A, percent of scheduled initially unfamiliar domain-rounds served successfully with any AI help the worker chose; percent of branch-rounds in which opening cash covered the minimum exploration purchase, before (solid) and after (dotted) the assistance chosen in the same round. Rows: all workers, then low, medium and high propensity (34, 36 and 36 worlds). Equal-world means over 106 complete worlds and both technologies; the shock and no-shock paths share rounds 1–5 and fork at round 6 (vertical line). Monetary affordability is not time for exploration (Table S-K4). Post-outcome exploratory description with no intervals; values and technology detail in Tables S-K1 to S-K3.

Cash accounts for much of what happened without the shock. Workers paid for exploration and assistance from the cash they held at the start of a round, before that round's earnings arrived, and at settlement they consumed half of what remained and carried the other half forward. The smallest exploration purchase, a quarter unit, cost 0.25 credits at the old price. Without the shock, opening cash covered it in 74 percent of the rounds before the shock but in only 393 of the 5,300 branch-rounds after it (7.4 percent), and 44 of the 58 no-shock branches that bought exploration and then stopped could not afford the minimum purchase in the round after their last one. In most rounds after round 5, therefore, a no-shock worker who bought no exploration had not chosen against it; she could not have paid for it. The price-plus-rebate package changed this arithmetic in two ways at once. From round 6 the price fell to a quarter, so the minimum purchase cost 0.0625 credits, and the rebate of 0.75 credits per round raised what each settlement carried forward. From round 7, opening cash in the shock branch never fell below 0.375 credits, six times the minimum purchase, which was affordable in 5,181 of the 5,300 branch-rounds (97.8 percent). The package thus removed the cash constraint on exploration as well as lowering its price, and the design does not separate the two. With the shock, almost every round without a purchase was one in which the worker could have paid for it. In 4,041 of those 5,038 rounds she still had the money after paying for assistance, but she had put three-quarters of the round into paid work, the most the time budget allows, and that leaves no time for an exploration purchase; another 33 were validator fallbacks after an invalid decision. Because she chose the allocation and the purchase together, these rounds describe the bundle she chose rather than a constraint on her (Table S-K4).

Where cash no longer constrained purchases, two further readings fit the later pattern, and the data do not choose between them. With the shock, 95 of the 113 branches that bought exploration and then stopped could still afford the minimum purchase, before and after their chosen assistance, in the round after their last one. One reading is late exploitation: a worker who already holds a usable card can serve its domain without buying more exploration. In the last five rounds of the shock branch, purchases fell from 0.0080 to 0.0047 units per round while successfully served unfamiliar work rose from 2.01 to 2.34 percent. The other reading is the finite horizon invoked in Section 7.1 of the 25 October draft: a domain explored in round 25 has few rounds left in which to pay. That reading fits the late stops, but it fits the fall without the shock less well, because that fall came in rounds 6–10, with twenty or more rounds remaining, when the share of rounds in which the minimum purchase was affordable dropped from 74 to 8 percent. Both patterns varied with propensity and service design (Tables S-K1 to S-K3). High-propensity workers in the shock branch bought between 0.0083 and 0.0139 units of exploration per round in rounds 6–25, against 0.0111 before the shock, and 0.0056 in rounds 26–30, while their successfully served unfamiliar work rose from 2.96 to 3.50 percent. That late fall was larger with the integrated adviser, whose high-propensity workers cut purchases from 0.0111 to 0.0014, than with the personalized adviser, from 0.0153 to 0.0097. Low-propensity workers in the shock branch bought at most half their early amount in every block after the shock, and medium-propensity workers with the integrated adviser bought more in rounds 26–30 than in any earlier block. The sequence of early purchases and later use therefore describes the average in the shock branch, not every group of workers.

Three questions remain open. The first is selection. Workers who bought exploration and later succeeded are not a random group, the analysis was specified after the outcomes were seen, and its successes are repeated events concentrated in a few branches and worlds: the ten most active branches account for 106 of the 417, and one world for 33. Because the two branches share rounds 1–5, the 44 purchases made in those rounds, and the stops that followed them, appear in both. The timing cannot separate what a purchase supplied from who chose to buy. The second is disappointing returns, which the data cannot assess directly. The worker saw her aggregate earnings and, every five rounds, an aggregate certification score, not her results domain by domain, and a one-time purchaser had no rounds between her only purchase and her stop in which a return could be observed. Among the 58 repeat purchasers, 24 had attempted a purchased domain without success before they stopped, 22 had succeeded at least once and 12 had made no attempt, so stopping followed both kinds of history. Few stops coincided with having nothing left to explore: of the 171 branches that stopped before round 30, five had filled all five card slots and none held a correct card for every initially unfamiliar domain. The third question is whether the recorded information supplied a given answer. The record shows which card was present before a success, not whether the card produced it, and a card may have been written from paid work on adjacent inputs or from a helper's method card rather than from purchased exploration.

Answering it would require a design that varies which cards are available when the worker answers.